Digital Electronics

Classroom Workbook

ELECTRONICS AND COMM. ENGINEERING

ELECTRICAL ENGINEERING

GATE/ESE

unacademy

Digital Electronics

Classroom Workbook

For ECE, EE

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GATE Syllabus

ECE

Number representations: binary, integer and floating-point- numbers. Combinatorial circuits: Boolean algebra, minimization of functions using Boolean identities and Karnaugh map, logic gates and their static CMOS implementations, arithmetic circuits, code converters, multiplexers, decoders. Sequential circuits: latches and flip-flops, counters, shift-registers, finite state machines, propagation delay, setup and hold time, critical path delay. Data converters: sample and hold circuits, ADCs and DACs. Semiconductor memories: ROM, SRAM, DRAM. Computer organization: Machine instructions and addressing modes, ALU, data-path and control unit, instruction pipelining.

EE

Combinatorial and sequential logic circuits, multiplexers, demultiplexers, A/D and D/A converters.

Contents

Chapters

1. Logic gates and Boolean Algebra
2. K-Maps
3. Number Systems and Codes
4. Combinational Circuits
5. Sequential Circuits
6. DACs and ADCs
7. CMOS Logic Family
8. Computer Organization

■■■■■

1 Logic Gates and Boolean Algebra

MCQ, MSQ and NAT

Q.1 Minimum number of 2-input NOR gates required to implement $Y = \bar{A}B + A\bar{B}\bar{C} + A\bar{B}$ is _____.

Q.2 The Boolean expression $X = (A + B)(B + C)(\bar{A} + C)$ is equivalent to-

(A)$AB + \bar{A}C$

(B)$(A + B)(\bar{A} + C)$

(C)$(\overline{A} + C)(B + C)$

(D)$AB + BC + \bar{A}C$

Q.3 The simplified form of the Boolean expression $F = (AB + \overline{\bar{A}BC})\overline{CD} + (A + \overline{\bar{B}\bar{C}})\bar{D}$ is-

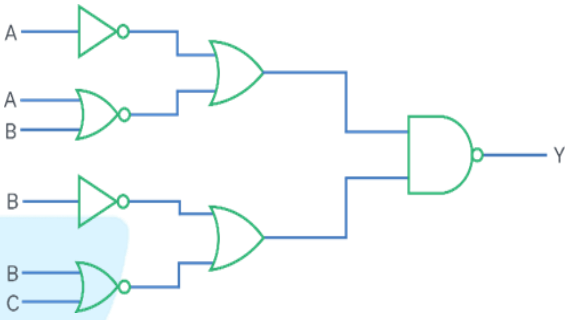
(A)$F = \overline{AB}$

(B)$F = \overline{\overline{A} + B}$

(C)$F = [\bar{C} + A\bar{D} + \bar{B}]$

(D)$F = \overline{CD}$

Q.4 The output expression of the given logic circuit is-



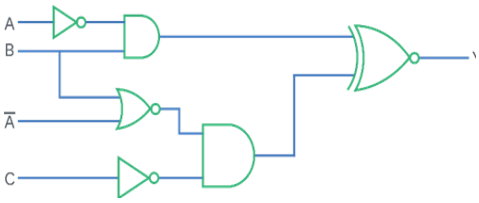
(A)$Y = \overline{\overline{A}B}$

(B)Y = A + B

(C)$Y = AB + \bar{C}$

(D)$Y = \overline{A + B}$

Q.5 The output of the circuit given below is-



(A)$Y = A \oplus B$

(B)$Y = \bar{A}B + A\bar{B}\bar{C}$

(C)Y = AB

(D)$Y = (A \odot B) + \bar{B}C$

Q.6 A factory has a security system as X, Y, and Z. An alarm is set off if two or more security conditions fail simultaneously. The logic of the system is-

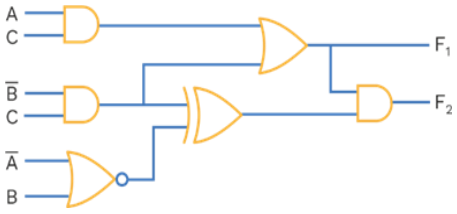
(A)X(Z + Y)

(B)Y(Z + X)

(C)XZ + YZ + XY

(D)$X\bar{Y}Z + \bar{X}YZ + XY\bar{Z}$

Q.7 Find the minterms of the logic circuit given below.



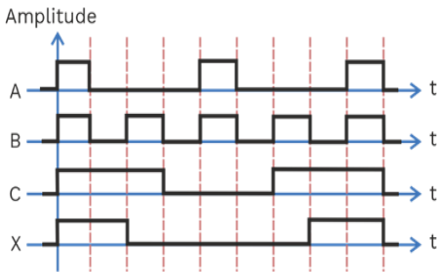
(A)$F_1 = \sum m(1,5,7)$

(B)$F_1 = \sum m(0,2,6,7)$

(C)$F_2 = \sum m(1,3)$

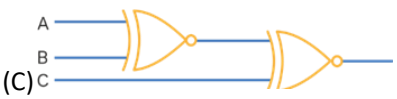
(D)$F_2 = \sum m(1)$

Q.8 A circuit consists of inputs A, B, and C and output X. The waveforms of inputs and output are given below. Select the corresponding logic circuit for which these waveforms are valid.



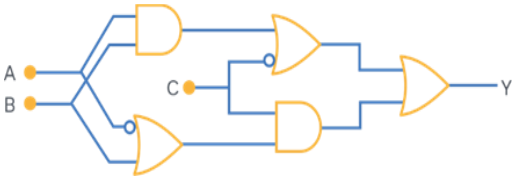
(A)


(B)


(C)


(D)


Q.9 For the logic circuit shown below, the output Y is-



(A)$AB\bar{C} + (\bar{A} + B)C$

(B)$\mathrm{A} + \overline{\mathrm{B}} + \mathrm{C}$

(C)$\overline{\mathrm{A}} + \mathrm{B} + \overline{\mathrm{C}}$

(D)$\overline{\mathrm{A}}\mathrm{BC} + (\mathrm{A} + \overline{\mathrm{B}})\overline{\mathrm{C}}$

Q.10 A Boolean function is given as-

$f(A,B,C,D) = ABCD + ABC + ABCD + AB$

The minimum number of 2-input NAND gates required for the implementation of the Boolean function are_____.

Q.11 Which of the following expression(s) hold(s) true?

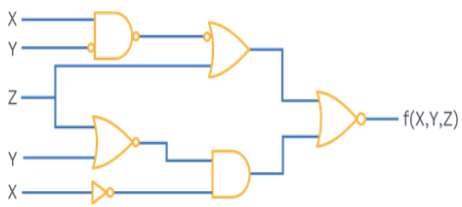
(A)$(X + \bar{Y}) \odot Y = \overline{XY}$

(B)$X \oplus Y \odot XY = \bar{X}\bar{Y}$

(C)$X \oplus (\bar{X} + Y) \odot Y = \overline{X + \bar{Y}}$

(D)$[\bar{X}Y \odot (X + \bar{Y})] + XY = XY$

Q.12 A digital circuit is shown in the figure below.



The output of the circuit f(X, Y , Z) = logic '1'. If the input X = 1, then Y and Z are-

(A)Y and Z can be '1' or '0'

(B)Y = '1' and Z = '0'

(C)Y and Z both should be '0'

(D)Cannot be determined

Q.13 A Boolean function is given as-

$f(W,X,Y,Z) = \overline{XY} + Y\bar{Z} + W\bar{Y}Z$

If the given Boolean function is realized using 8 × 1 MUX using W as its select line, then-

(A)minimized expression of f is f = X + Y + Z

(B)input at I_3 = 1 and I_7 = 0

(C)minimized expressed of f is $\mathrm{f} = \overline{\mathrm{X}} + \overline{\mathrm{Y}} + \overline{\mathrm{Z}}$

(D)input at I_0 = I_7 = 0

Q.14 The correct equivalent expression(s) of $\mathrm{F} = \mathrm{X} \oplus \mathrm{Y} \oplus \mathrm{XY}$ is/are

(A)F = X + Y

(B)$F = \overline{\bar{X}\bar{Y}}$

(C)$\mathrm{F} = \mathrm{X} \oplus \overline{\mathrm{Y}}$

(D)$F = \overline{\bar{X}Y}$

Q.15 The minimum number of 2-input NAND gates required to realize the function

$f(A,B,C,D) = \Sigma m(0,1,4,5,10,11,14,15) + \sum d(8,9,12,13)$

is_____.

Q.16 Which of the following is the simplified boolean algebra version of $(P+Q)\cdot(P+R)$?

(A)$P + Q \cdot R$

(B)$P + P \cdot (Q + R)$

(C)P

(D)$P.(1 + Q.R)$

Q.17 The minimized POS expression for F (a, b, c, d) = Σm(0, 1, 3, 5, 7, 8, 9, 11) is-

(A)$\bar{\mathrm{b}}\bar{\mathrm{c}} + \mathrm{ad} + \bar{\mathrm{b}}\mathrm{d}$

(B)$(\bar{b} + d) \cdot (\bar{a} + \bar{b}) \cdot (\bar{c} + d)$

(C)$(\bar{b} + \bar{c})(a + d)(\bar{b} + d)$

(D)$\bar{\mathrm{b}}\mathrm{d} + \bar{\mathrm{a}}\bar{\mathrm{b}} + \bar{\mathrm{c}}\mathrm{d}$

Q.18 The minimum number of 2 input NAND gates required for the realization of the given expression $f(AB) = \overline{\overline{A\bar{B}} \cdot \overline{\bar{A}B}}$ is _____.

Q.19 Consider the following expression,

$f(A,B,C,D) = AD + ABCD + ACD + \bar{A}B + A\bar{C}D + \bar{A}\bar{B} + \bar{A}(\bar{B} + A)$

The minimized expression of $f(A, B, C, D)$ is

(A)AD

(B)$\bar{A} + D$

(C)$A + B\bar{C}$

(D)$A\bar{D} + B\bar{C}$

Q.20 The given Boolean function $\mathrm{F} = \overline{\mathrm{A}} \oplus \overline{\mathrm{B}} \oplus \overline{\mathrm{C}}$ is equivalent to which of following expression?

(A)$F = A \oplus B \oplus C$

(B)$\mathrm{F} = \mathrm{A} \odot \mathrm{B} \odot \mathrm{C}$

(C)$\mathrm{F} = \mathrm{A} \oplus \overline{\mathrm{B}} \oplus \overline{\mathrm{C}}$

(D)$F = A \oplus B \odot C$

Q.21 The simplified form of the given boolean function is

$\mathrm{F(A,B,C,D)} = \bar{\mathrm{A}}\bar{\mathrm{B}} + \mathrm{AB}\bar{\mathrm{C}}\mathrm{D} + \bar{\mathrm{D}}\mathrm{BA} + \mathrm{BC}\bar{\mathrm{D}} + \bar{\mathrm{C}}\mathrm{D}$

(A)$F(A,B,C,D) = \bar{C}D + \bar{A}\bar{B} + BC\bar{D} + \bar{A}B\bar{C}$

(B)$F(A,B,C,D) = (C+D)(A+B)(\bar{B}+\bar{C}+D)(\bar{A}+\bar{B}+C)$

(C) $F(A,B,C,D)=\overline{\bar{C}D+\bar{A}\bar{B}+BC\bar{D}+AB\bar{C}}$

(D) $F(A,B,C,D)=\overline{(C+D)(A+B)(\bar{B}+\bar{C}+D)(\bar{A}+\bar{B}+C)}$

Q.22 For the following expression, which of the conclusion/conclusions can be drawn

$F(x,y,z)=\bar{x}y+xz\bar{y}+xzy+\bar{x}yz+\overline{(\bar{x}+\bar{y}+z)\cdot(x+\bar{y}+\bar{z})}$

(A)$f=x\bar{y}+\bar{x}z$

(B)No of literals are 4

(C)No of literals are 3

(D)f = y + xz

Q.23 The number count of 2- input NOR gate required to realize $y_1=\overline{x_1+x_2+x_3+x_4}$ is n_1 and the number count of 2- input NAND gate required to realize $y_2=\overline{x_1x_2x_3x_4}$ is n_2 Then

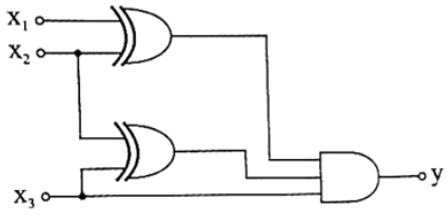
(A)$n_1=n_2=5$

(B)$n_1=n_2=4$

(C)$n_1=n_2=6$

(D)$n_1=n_2=3$

Q.24 The digital circuit shown below, is observed to generate output y=1. Which one of the following input condition does so?



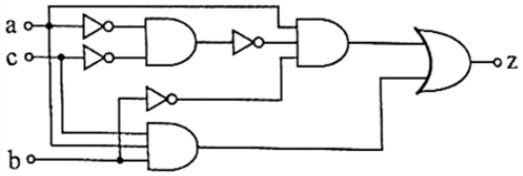
(A) $x_1=x_3=1, x_2=0$

(B) $x_2=x_3=1, x_1=0$

(C) $x_1=x_2=x_3=0$

(D) $x_1=x_2=x_3=1$

Q.25 In the logic circuit shown below, how many AND, how many OR and how many NOT gates are Redundant?



(A)2 AND gates and 3 NOT gates

(B)2 AND gates and 2 NOT gates

(C)2 AND gates, 2 NOT gate and 1 OR gate

(D)3 AND gates and 3 NOT gates

Q.26 The minimum number of 2- input NAND gates required to realize boolean function (F) = (W + X)(Y + Z) is_____.

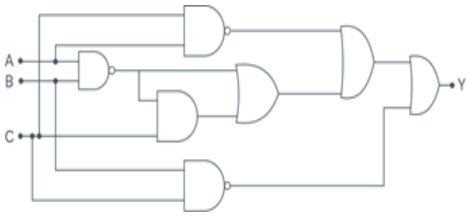
(A)7

(B)5

(C)6

(D)8

Q.27 The output of the circuit shown in the figure is-



(A)$\overline{AB}+\overline{AC}+\overline{BC}$

(B)$\overline{AB}+\overline{AC}+\overline{BC}+\overline{ABC}$

(C)$A+B+C$

(D)$\overline{ABC}$

Q.28 The simplified form of Boolean expression,

$y = (P + \bar{Q} + \bar{R}) \cdot (P + \bar{Q} + R)$ is

(A)$\overline{\mathrm{P}} \cdot \mathrm{Q}$

(B)P+Q

(C)$\mathrm{P} + \overline{\mathrm{Q}}$

(D)$\mathrm{P}\overline{\mathrm{Q}}$

Q.29 The Boolean expression

$y = \bar{A}\bar{B}\bar{C}D + \bar{A}B\bar{C}\bar{D} + \bar{A}BC\bar{D} + AB\bar{C}\bar{D} + A\bar{B}\bar{C}D$

can be minimized to

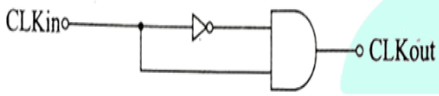
(A)$\mathrm{BCD} + \mathrm{BCD} + \mathrm{ABD}$

(B)$ABCD + \bar{A}B\bar{C} + A\bar{C}D$

(C)$\mathrm{BC\bar{D}} + \mathrm{\bar{A}\bar{B}\bar{C}D} + \mathrm{A\bar{B}\bar{C}D}$

(D)$\mathrm{\bar{A}BC\bar{D}} + \mathrm{B\bar{C}\bar{D}} + \mathrm{\bar{A}B\bar{D}}$

Q.30 An edge detector circuit used in edge triggered flops, is shown below, the inverter has delay of 10 ns and AND gate has zero delay. Then consider the following statements.



I. The frequency of CLK_{in} will be equal to frequency of CLK_{out}

II.CLK_{out} will have narrow pulses of 10 ns width

III.The narrow pulses of CLK_{out} will occur at NGT (Negative Going Transition) of CLK_{in} of these, the Correct statements are

(A)I and II

(B)I and III

(C)II and III

(D)I, II and III

Q.31 Minimum number of 2-input NOR gates required to realize

$F = \bar{A}B + XY$ is:

(A)6

(B)4

(C)2

(D)5

Q.32 The given Boolean function

$F = \bar{A} \oplus \bar{B} \oplus \bar{C}$ is equivalent to which of following expression.

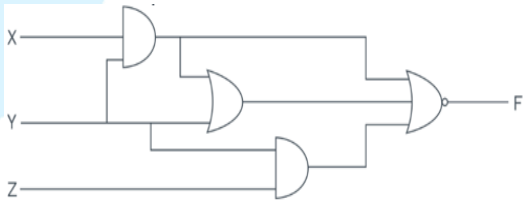
(A)$\mathrm{F} = \mathrm{A} \oplus \mathrm{B} \oplus \mathrm{C}$

(B)$\mathrm{F} = \mathrm{A} \odot \mathrm{B} \odot \mathrm{C}$

(C)$\mathrm{F} = \mathrm{A} \oplus \overline{\mathrm{B}} \oplus \overline{\mathrm{C}}$

(D)$\mathrm{F} = \mathrm{A} \oplus \mathrm{B} \odot \mathrm{C}$

Q.33 In the given circuit, the output F is-



(A)$F = \overline{XY + YZ + Y}$

(B)$F = \overline{XY + Y}$

(C)$F = \overline{Y}$

(D)All of the above

Q.34 The truth table of a circuit is shown below:

Inputs			Outputs
A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	X
0	1	1	X
1	0	0	1
1	0	1	X
1	1	0	1
1	1	1	1

(I) $\mathrm{Y(A, B, C)} = \mathrm{B} + \overline{\mathrm{C}}$

(ii) $\mathrm{Y(A, B, C)} = \mathrm{A} + \overline{\mathrm{C}}$

The Boolean expression for Y is:

(A)Only (i)

(B)Only (ii)

(C)Both (i) and (ii)

(D)Neither (i) nor (ii)

Q.35 The minimum number of NAND gates required to implement $F = P + P\,\overline{Q} + P\,\overline{Q}\,R$ is -

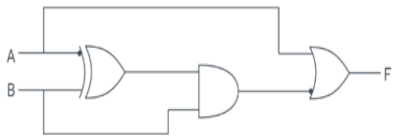
(A)3

(B)0

(C)1

(D)5

Q.36 A circuit is given below:



The output F is-

(A)$\overline{B}$

(B)0

(C)1

(D)$\bar{A} + \bar{B}$

Q.37 Find the dual of the Boolean expression-

$A\bar{B} + \bar{C}\bar{D} + (D + A) = (\bar{A} + D) + CD$

(A)$\overline{\mathrm{A}}\mathrm{B} + \mathrm{CD} + (\overline{\mathrm{C}} + \overline{\mathrm{A}}) = (\mathrm{A} + \overline{\mathrm{D}}) + \overline{\mathrm{CD}}$

(B)$A\bar{B} \cdot (\bar{C}\bar{D}) \cdot (DA) = (\bar{A} \cdot D) \cdot (C + D)$

(C)$(\bar{A}+B)(C+D)(\overline{\mathrm{D}}\ \overline{\mathrm{A}}) = (\mathrm{A}\overline{\mathrm{D}}) \cdot (\overline{\mathrm{C}} + \overline{\mathrm{D}})$

(D)$\overline{(\bar{A}+B)(C+D)\bar{D}\bar{A}} = \overline{(A\bar{D}) \cdot \bar{C} + \bar{D})}$

Q.38 Determine the minimized expression of given Boolean function –

$Y(A, B, C) = \bar{A}\bar{C} + \bar{B}\bar{C} + B\bar{C} + ABC$

(A)$Y(A, B, C) = ABC$

(B)$Y(A, B, C) = \bar{A}B + C$

(C)$Y(A, B, C) = \bar{C} + AB$

(D)$Y(A, B, C) = \bar{A}\bar{B} + C$

Q.39 A bulb in a staircase has two switches, one switch being at the ground floor and the other one at the first floor. The bulb can be turned ON and also can be turned OFF by any one of the switches irrespective of the state of the other switch. The logic of switching of the bulb resembles

(A) an AND gate

(B) an OR gate

(C) an XOR gate

(D) a NAND gate

Q.40 The Boolean expression $(\mathrm{X} + \mathrm{Y})(\mathrm{X} + \overline{\mathrm{Y}}) + \overline{(\mathrm{X}\overline{\mathrm{Y}}) + \overline{\mathrm{X}}}$ simplifies to

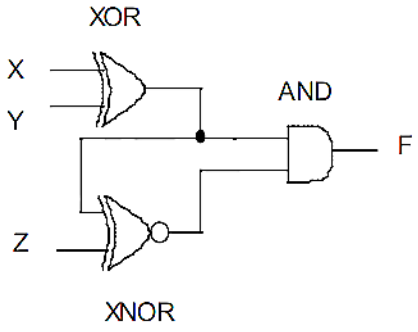
(A) X

(B) Y

(C) XY

(D) X + Y

Q.41 The output F in the digital logic circuit shown in the figure is



(A) $\mathrm{F} = \overline{\mathrm{X}}\mathrm{YZ} + \mathrm{X}\overline{\mathrm{Y}}\mathrm{Z}$

(B) $\mathrm{F} = \overline{\mathrm{X}}\mathrm{Y}\overline{\mathrm{Z}} + \mathrm{X}\overline{\mathrm{Y}}\overline{\mathrm{Z}}$

(C) $\mathrm{F} = \overline{\mathrm{X}}\overline{\mathrm{Y}}\mathrm{Z} + \mathrm{XYZ}$

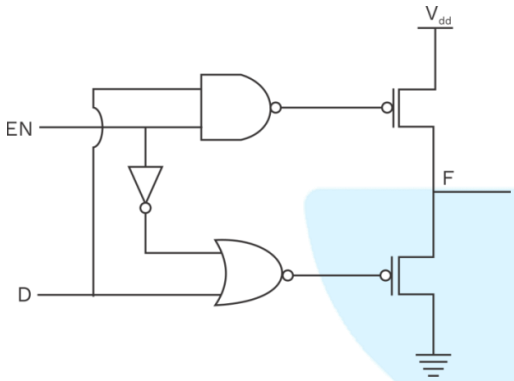
(D) $F = \overline{X}\overline{Y}\overline{Z} + XYZ$

Q.42 For an *n*-variable Boolean function, the maximum number of prime implicants is

(A) $2(n-1)$ (B) $n/2$

(C) 2^n (D) $2^{(n-1)}$

Q.43 In the circuit shown, what are the values of F for EN = 0 and EN = 1, respectively?



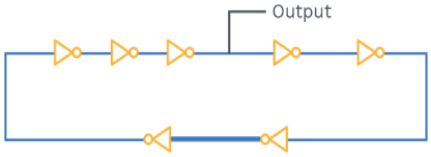
(A) 0 and 1

(B) Hi – Z and $\overline{D}$

(C) 0 and D

(D) Hi – Z and D

Q.44 A circuit is shown in the figure below.



The inverters are identical in the ring oscillator. The output waveform has a frequency of 2.5 MHz. The propagation delay of the individual inverters is_____ nsec. (Up to 2 decimal places)

Answer Key

Q.1 Ans. [5]

Q.2 Ans. [B]

Q.3 Ans. [D]

Q.4 Ans. [B]

Q.5 Ans. [D]

Q.6 Ans. [C]

Q.7 Ans. [A ,D]

Q.8 Ans. [C]

Q.9 Ans. [C]

Q.10 Ans. [3]

Q.11 Ans. [B ,C ,D]

Q.12 Ans. [B]

Q.13 Ans. [B ,C]

Q.14 Ans. [A ,B]

Q.15 Ans. [2]

Q.16 Ans. [A]

Q.17 Ans. [B]

Q.18 Ans. [4]

Q.19 Ans. [B]

Q.20 Ans. [D]

Q.21 Ans [D]

Q.22 Ans. [C ,D]

Q.23 Ans. [A]

Q.24 Ans. [A]

Q.25 Ans. [A]

Q.26 Ans [C]

Q.27 Ans. [A ,B ,C ,D]

Q.28 Ans. [C]

Q.29 Ans. [A]

Q.30 Ans. [A]

Q.31 Ans [D]

Q.32 Ans [D]

Q.33 Ans. [D]

Q.34 Ans. [C]

Q.35 Ans. [B]

Q.36 Ans. [C]

Q.37 Ans. [C]

Q.38 Ans. [C]

Q.39 Ans. [C]

Q.40 Ans. [A]

Q.41 Ans. [A]

Q.42 Ans. [D]

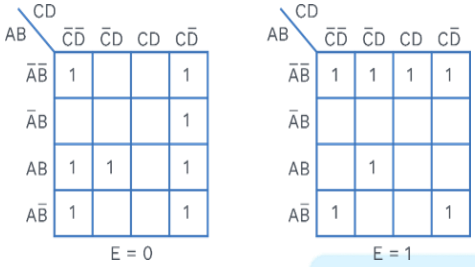
Q.43 Ans. [D]

Q.44 Ans. [28.57]

2 K-Maps

MCQ, MSQ and NAT

Q.1 A five variable K-map is shown in the figure below.



The minimized expression for the given K-map is-

(A) $\bar{B}\bar{D}+C\bar{D}E+\bar{A}\bar{B}E+AB\bar{C}\bar{E}+A\bar{D}\bar{E}+AB\bar{C}DE$

(B) $\bar{B}\bar{D}+C\bar{D}\bar{E}+A\bar{D}\bar{E}+AB\bar{C}D+\bar{A}\bar{B}E$

(C) $\bar{B}\bar{D}E+C\bar{D}E+\bar{A}\bar{B}E+AB\bar{C}\bar{E}+A\bar{D}\bar{E}+AB\bar{C}DE+\bar{B}\bar{D}\bar{E}$

(D) $\bar{B}\bar{D}+C\bar{D}\bar{E}+A\bar{D}\bar{E}+A\bar{B}E+AB\bar{C}\bar{E}+AB\bar{C}D$

Q.2 The digital inputs to a system are W, X, Y, and Z, and the output is given as F. The output F is defined as-

$F = m_0 + m_2 + m_5 + m_7 + m_{10}$

The output F has don't care for m_8, m_{13}, and m_{15}.

The simplified expression of F is-

(A) $X \odot Z$

(B) $X \oplus Z$

(C) X

(D) Z

Q.3 A four variable Boolean function is defined as-

f(A, B, C, D) = π(2, 3, 4, 5, 6, 7, 10, 11)

The minimum number of 2-input NOR gates required to implement the Boolean function $\overline{f(A, B, C, D)}$ are _____ .

(Assume that the inputs are not available in complement form)

Q.4 Consider the Boolean function-

$$f = \sum m(0, 1, 3, 5, 7, 8) + \sum d(2, 10, 11, 15)$$

The number of literals is POS expression are_____.

Q.5 Simplify the given Boolean function,

F(A, B, C, D) = πM(4, 5, 6, 7, 8, 12) + d(0, 1, 2, 3, 9, 11)

(A) $F(A, B, C, D) = \overline{\bar{A} + \bar{C}\bar{D}}$

(B) $F(A, B, C, D) = \bar{A}(C + D)$

(C) $F(A, B, C, D) = \bar{A} + A\bar{C}\bar{D}$

(D) $F(A, B, C, D) = \overline{A}(\overline{C} + \overline{D})$

Q.6 Simplify the given logic function using k-map

F(P, Q, R, S, T) = Σm(0, 2, 3, 7, 11, 13, 15, 16, 18, 19, 21, 23, 27, 31)

(A) $F = RS + RS\bar{T} + \bar{R}\bar{S}\bar{T} + PQST$

(B) $F = RS + \bar{P}Q\bar{S} + PQST + \bar{P}QS\bar{T}$

(C) $F = RS + \bar{P}\bar{Q}\bar{S} + PQS\bar{T} + \bar{P}QST$

(D) $F = RS + \bar{P}\bar{Q}\bar{S} + PQ\bar{S} + QST$

Q.7 Minimize the given function,

$$F(A, B, C, D) = \Sigma m(0, 2, 3, 5, 9, 12, 14) + \Sigma d(1, 10)$$

(A) $F(A,B,C,D) = \bar{A}\bar{B} + AB\bar{D} + \bar{A}\bar{C}D + \bar{B}\bar{C}D$

(B) $F(A,B,C,D) = (\overline{A+B}) + (\overline{\bar{A}+\bar{B}+\bar{D}}) + (\overline{A+C+\bar{D}}) + (\overline{B+C+\bar{D}})$

(C) $F(A,B,C,D) = \bar{A}\bar{B} + AC\bar{D} + \bar{A}B\bar{C}D + \bar{B}\bar{C}D$

(D) $F(A,B,C,D) = \overline{(\bar{A}+\bar{B})+(A+\bar{B}+D)+(A+C+\bar{D})+\bar{B}+\bar{C}+D}$

Q.8 Consider the following statements.

I. The minimal expression for $f_1(a,b,c,d) = (\bar{a}bc+d)(\bar{a}d+b\bar{c})$ is $f_1 = d(\bar{a}+\bar{c})(\bar{a}+b)$

II. The number count of max terms and min terms in the expression $f_2(a,b,c,d,e) = 1$, are 0 and 32 respectively.

III. The number count of max terms and min terms in the expression $f_3(x_1,x_2,x_3,x_4) = 0$, are 1 6 and 0 respectively.

Of these, the correct statements are

(A)I is correct

(B)II is correct

(C)III is correct

(D)Only I & II are correct

Q.9 The minimized SOP form of the K-map given below is-

CD / AB	00	01	11	10
00	×	1	×	1
01	1	×	×	1
11	0	0	1	0
10	0	1	1	0

(A) $AC\bar{D} + A\bar{B}D + \bar{A}B\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}C\bar{D}$

(B) $AC\bar{D} + A\bar{B}D + \bar{A}B\bar{D} + \bar{A}C\bar{D} + \bar{B}\bar{C}D$

(C) $\bar{A} + AC\bar{D} + \bar{B}D$

(D) $\bar{A} + C\bar{D} + \bar{B}D$

Q.10 In the sum of products function $f(X,Y,Z) = \sum(2,3,4,5)$, the prime implicants are

(A) $\bar{X}Y, X\bar{Y}$

(B) $\bar{X}Y, X\bar{Y}\bar{Z}, X\bar{Y}Z$

(C) $\bar{X}Y\bar{Z}, \bar{X}YZ, X\bar{Y}$

(D) $\bar{X}Y\bar{Z}, \bar{X}YZ, X\bar{Y}\bar{Z}, X\bar{Y}Z$

Q.11 Consider the Boolean function,

F(w,x,y,z)

$= wy + xy + \bar{w}xyz + \bar{w}\bar{x}y + xz + \bar{x}\bar{y}\bar{z}$.

Which one of the following is the complete set of essential prime implicants?

(A) $w, y, xz, \bar{x}\bar{z}$

(B) w, y, xz

(C) $y, \bar{x}\bar{y}\bar{z}$

(D) $y, xz, \bar{x}\bar{z}$

Answer Key

Q.1 Ans [B]

Q.2 Ans. [A]

Q.3 Ans [5]

Q.4 Ans. [4]

Q.5 Ans. [A]

Q.6 Ans [C]

Q.7 Ans. [A ,B]

Q.8 Ans. [A ,B ,C]

Q.9 Ans. [D]

Q.10 Ans. [A]

Q.11 Ans. [D]

3 Number Systems and Codes

MCQ, MSQ and NAT

Q.1 The base (b) of the number system for which the arithmetic operation $\sqrt{(51)_b} = 6$ is valid, is-

(A)6 (B)7

(C)8 (D)5

Q.2 If $(756.603)_8 = (X)_{16}$, then the value of X is approximately equal to-

(A) 1EE.024

(B) A26.ACC

(C) 1EE.C18

(D) 127.C2A

Q.3 The 15's complement of the hexadecimal number $(F23.2AE)_{16}$ is-

(A)$(A21.325)_{16}$

(B)$(2AE.F23)_{16}$

(C)$(DC.D52)_{16}$

(D)$(DC.D51)_{16}$

Q.4 The range of signed decimal numbers that can be represented by 9-bit 1's complement number is-

(A)–127 to +128

(B)–255 to +256

(C)–255 to +255

(D)–1023 to +1023

Q.5 If the 16's complement of Y is $(AF82.AA)_{16}$, then the subtraction of Y from $(4AB.68)_{16}$ is-

(A)$(120A.2E)_{16}$

(B)$(507D.56)_{16}$

(C)$(B42E.12)_{16}$

(D)$(BB62.1A)_{16}$

Q.6 The 9's complement of $(231)_6$ is_____.

Q.7 Find the addition of $(374.26)_8$ and $(1110100.0100111)_2$

(A)$(236.025)_8$

(B)$(134.261)_8$

(C)$(560.514)_8$

(D)$(601.006)_8$

Q.8 The decimal sum of gray codes which are converted from decimal equivalent (0 – 8) is_____. (Rounded off to the nearest integer).

Q.9 Find out the missing number (x) in the series. $(0111)_2$, $(42)_3$, $(51)_4$, $(34)_8$, $(x)_{16}$. (Rounded off to the nearest integer)

Q.10 The product of two signed 2's complement binary numbers X = 11101001 and Y = 11111110 is

(A)00101110

(B)00111001

(C)01011100

(D)10111010

Q.11 If X = 512 × 5 + 64 × 2 + 12, then number of 1's present in the binary equivalent of X is_____.

Q.12 Consider the relation $\left.\frac{228}{21}\right|_{basex} = (11)_{basex}$. The value of base x is_____.

Q.13 Consider the relation $(323)_5=(y8)_x$. If x, y $\in$(I), how many combinations of (x, y) are possible?

Q..14 If $(42)_y + (33)_y = (130)_y$, then, the value of base 'y' is-

(A)5 (B)6

(C)8 (D)None

Q.15 The r's complement of an n-digit decimal number N in base r is defined for all values of N except for N = 0. If the given number is $(247)_9$, then its complement is equal to $(_____)_9$.

Q.16 Consider the following statements:

I. The number 11101.11 on base 2 is equivalent to 1 D.3 on base 16.

II. The number 1001.1 on base 2 is equivalent to 11.4 on base 8.

III. The number 112 on base 4 is equivalent to 211 on base 3.

IV. The number 214 on base 5 is equivalent to 3 B on base 16.

Among these statements, the correct statement(s) is/are

(A)I is correct

(B)II is correct

(C)III is correct

(D)IV is correct

Q.17 Given $(135)_b + (144)_b = (323)_b$ where subscript b denotes the base on which numbers are expressed. What is value of b_____ ?(in integer)

Q.18 When the value $(334)_8$ is divided by 28 H, the remainder is

(A)$(15)_{16}$

(B)$(25)_7$

(C)$(22)_9$

(D)$(50)_4$

Q.19 If $(\sqrt{\sqrt{69}})_a = (0011)_2$, then the value of base 'a' is _____.

Q.20 If $(36.45)_7 = (A)_5$, then the value of A is _____. (up to three decimal places)

Q.21 In which number system, the operation $\frac{54}{4} = 13$ is-

(A)Binary number system

(B)Octal Number system

(C)Decimal Number system

(D)Hexadecimal Number system

Q.22 Find the binary and decimal equivalent of $(62.32)_8$

(A)110010.01101 & 50.40625 respectively

(B)110010.011011& 55.625 respectively

(C)101110.101001 & 42.125 respectively

(D)110010.01101 & 62.25 respectively

Q.23 $\text{If } (65)_8 \times (15)_{10} = (XYZ)_{16}, \text{ then :}$

(A)X = 0, Y = A and Z = 8

(B)X = 2, Y = 9 and Z = B

(C)X = 2, Y = D and Z = B

(D)X = 0, Y = 2 and Z = 8

Q.24 The product of two signed 2's complement binary numbers X = 10010101 and Y = 11111110 is-

(A)11010110

(B)00111001

(C)0101 1100

(D)10111010

Q.25 The number of bytes required to represent the decimal number 1856357 in packed BCD (Binary Coded Decimal) form is _____.

Q.26 P, Qand R are the decimal integers corresponding to the 4-bit binary number 1100 considered in signed magnitude, 1's complement and 2's complement representations, respectively. The 6-bit 2's complement representation of (P + Q + R) is:

(A) 110101

(B) 110010

(C) 111101

(D) 111001

Q.27 If $(1235)_x = (3033)_y$, where x and y indicate the bases of the corresponding numbers, then

(A) x = 7 and y = 5

(B) x = 8 and y = 6

(C) x = 6 and y = 4

(D) x = 9 and y = 7

Q.28 The decimal sum of gray codes which are converted from decimal equivalent (0 – 8) is_____. (Rounded off to the nearest integer).

Answer Key

Q.1 Ans. [B]

Q.2 Ans. [C]

Q.3 Ans. [D]

Q.4 Ans. [C]

Q.5 Ans. [C]

Q.6 Ans. [8]

Q.7 Ans. [C]

Q.8 Ans. [40]

Q.9 Ans. [23]

Q.10 Ans. [A]

Q.11 Ans. [5]

Q.12 Ans. [7]

Q.13 Ans. [5]

Q..14 Ans. [A]

Q.15 Ans. [642]

Q.16 Ans. [B ,C ,D]

Q.17 Ans. [6]

Q.18 Ans. [C ,D]

Q.19 Ans. [12]

Q.20 Ans. [102.314]

Q.21 Ans [B]

Q.22 Ans. [A]

Q.23 Ans. [C]

Q.24 Ans. [A]

Q.25 Ans. [4]

Q.26 Ans. [A]

Q.27 Ans. [B]

Q.28 Ans. [40]

4 Combinational Circuits

MCQ, MSQ and NAT

Q.1 An 8×1 multiplexer is used to implement the Boolean function $F(A,B,C,D) = \bar{A}\bar{B}\bar{D} + ABC + \bar{B}CD + \bar{A}CD$. If B, C, and D are used as select lines, then-

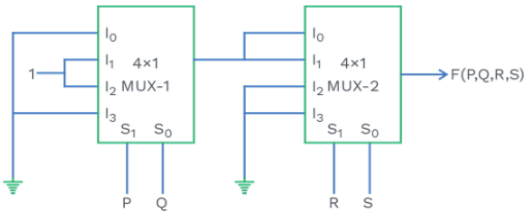
(A)$I_2 = \overline{A}$

(B)$I_4 + I_6 = A$

(C)$I_0 + I_7 = 1$

(D)$I_3 = 0$

Q.2 A four-variable function is realized using 4 × 1 multiplexers as shown in the figure below.



The minimized expression for F (P, Q, R, S) is-

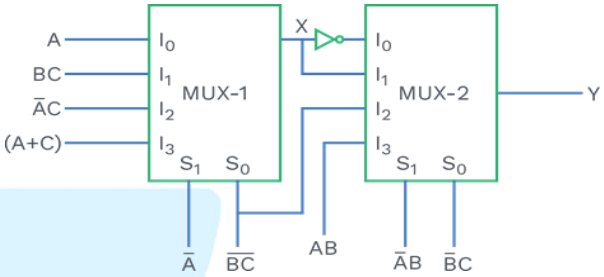
(A)$(P \oplus Q)(R \oplus S)$

(B)$\bar{R}(P \oplus Q)$

(C)1

(D)$P \oplus Q$

Q.3 The outputs (X and Y) of the given multiplexer circuit are-



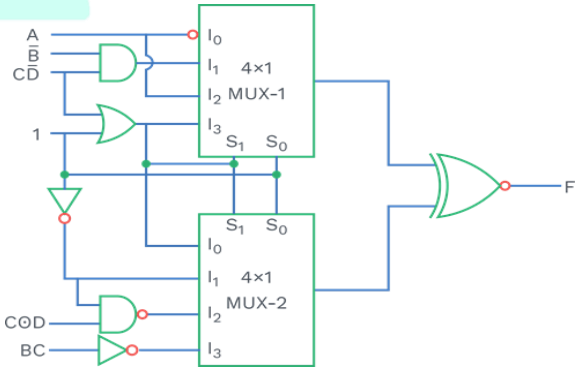
(A)$Y = \bar{A}\bar{B} + \bar{C}$

(B)$X = C(\bar{A} + B)$

(C)$Y = AB\bar{C} + \bar{A}(B \oplus C)$

(D)$X = B\bar{C}$

Q.4 The output (F) of the given combinational circuit is-



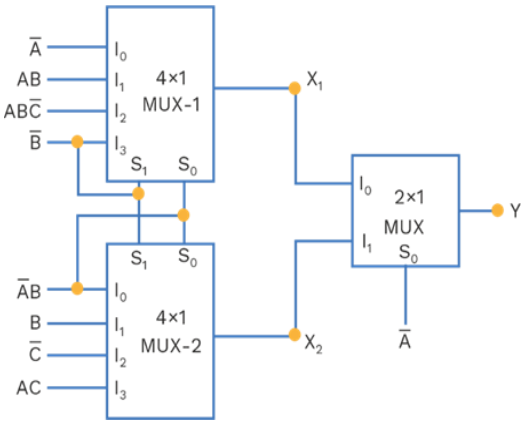
(A)$F = \bar{B}C + D$

(B)$F = 1 + C\bar{D}$

(C)F = 1

(D)$F = \overline{BC}$

Q.5 A combinational circuit is shown in the figure below. The output (Y) is -



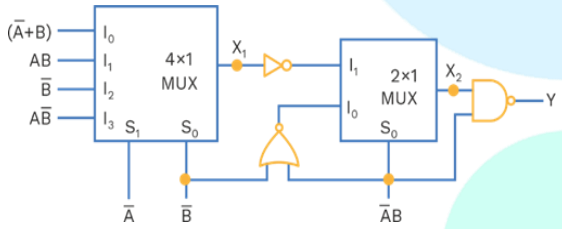
(A)$Y = \overline{A} + B$

(B)$Y = \bar{B}C + AB$

(C)$Y = \bar{A}B + \bar{A}\bar{C}$

(D)$Y = A\bar{B}C + AC$

Q.6 Consider the two cascaded multiplexers as shown in the figure below. Choose the correct option(s).



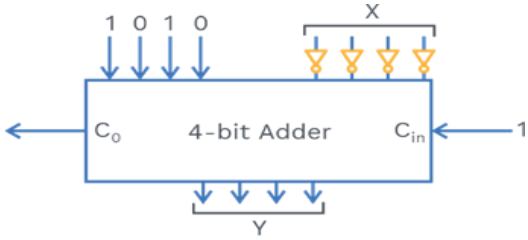
(A)$Y = A + \bar{B}$

(B)$Y = \overline{A\bar{B}}$

(C)$X_1 = \bar{A}B$

(D)$\bar{X}_1 = \bar{A} + \bar{B}$

Q.7 An adder circuit is shown below.



The input X to the 4-bit adder is a BCD number.

The circuit-

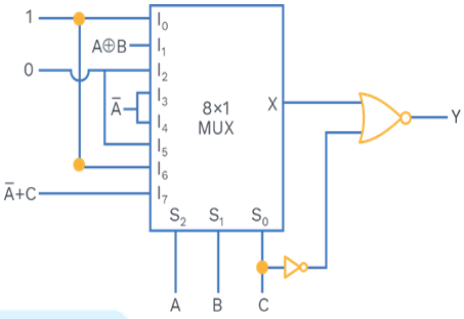
(A)Converts X to Gray code

(B)Provides 9's complement of X

(C)Converts X to 2-4-2-1 code

(D)Provides 10's complement of X

Q.8 A combinational circuit is shown in the figure below.



The output Y is-

(A)$\overline{B}C$

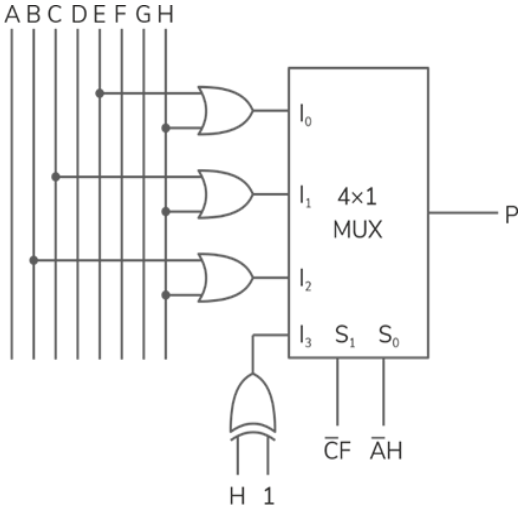
(B)$A\bar{B}C + ABC$

(C)$A\bar{B}C$

(D)$\bar{A}\bar{B}\bar{C} + BC + AB\bar{C}$

Q.9 An 8 × 3 Encoder circuit is cascaded with 4 × 1 multiplexer as shown below. Find the output (P) of the circuit if A = C = D = X, B = E = Y and F = G = H = Z.

(Assume X, Y, and Z are also available in complemented form).



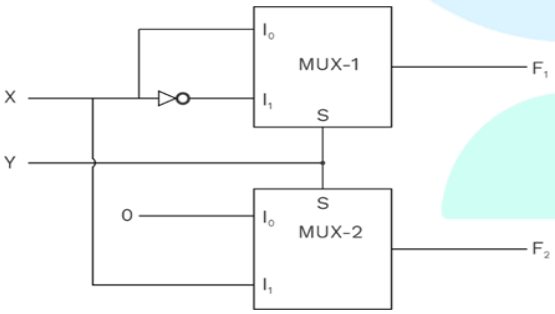
(A)$P = X\bar{Z}$

(B)$P = (X \oplus Y)Z$

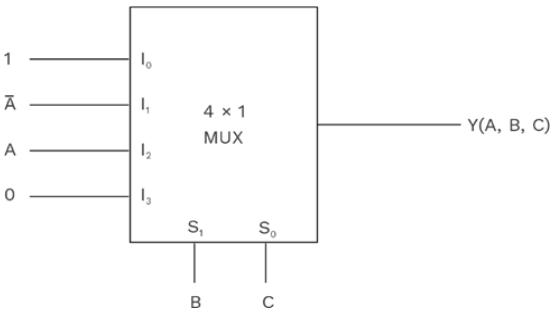
(C)$P(X, Y, Z) = Y + XZ$

(D)$P = \bar{X}Y + \bar{Z}$

Q.10 The minimum number of NAND gates required to implement the given combinational circuit is_____.



Q.11 A 4 × 1 MUX is used to implement a 3-input Boolean function as shown in the figure. The Boolean function Y(A, B, C) is



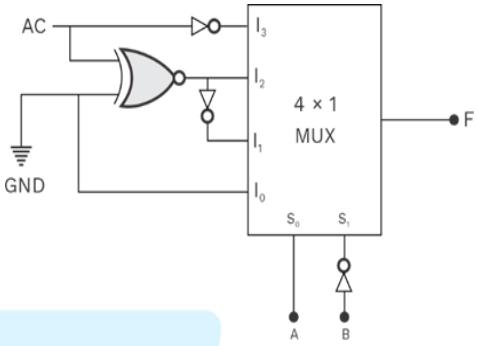
(A)Y(A,B,C) = Σm(0,1,4,6)

(B)Y(A, B, C) = πM(0,1,4,6)

(C)Y(A,B,C) = Σm(2, 3, 5, 7)

(D)Y(A, B, C) = πM(2, 3, 5, 7)

Q.12 For the combinational circuit shown below, find the output F. (Assume ground = logic '0')



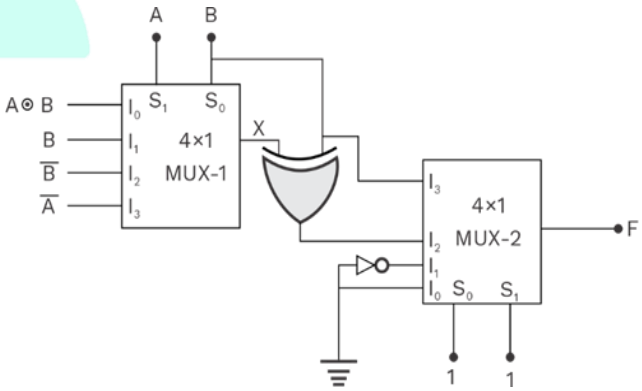
(A)$F(A, B, C) = ABC + \bar{A}\bar{B} + \bar{B}\bar{C}$

(B)$\mathrm{F(A, B, C)} = (\mathrm{A} \odot \mathrm{C})$

(C)$F(A, B, C) = (\overline{\bar{A} + C})$

(D)$F(A, B, C) = (\overline{\bar{A} + \bar{C}})$

Q.13 Consider the circuit shown in the figure below.



The Boolean expression(s) for the given circuit is/are-

(Where ground implies the logic '0')

(A)Output F = B

(B)$x = \overline{\mathrm{AB}}$

(C)$x = \bar{A}\bar{B} + (A \oplus B)$

(D)Output $F = (A \oplus B)$

Q.14 A $2^n \times 1$ multiplexer is designed using 2×1 multiplexers only. Find the number of 2×1 multiplexers required for the construction of $2^n \times 1$ multiplexer.

(A)2^{n-1} (B)2^n

(C)2^{2n} (D)2^n-1

Q.15 A decoder with 'p' input lines and 'q' output lines is used to address the memory of 2 kB RAM uniquely. Then the minimum value of 'p + q' is _____. (Rounded off to the nearest integer).

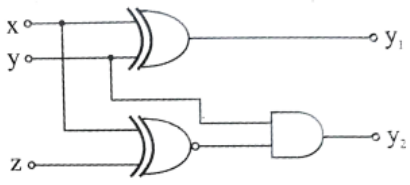
Q.16 Number of 2×4 decoders needed to implement 4×16 decoder without extra hardware is-

(A)4 (B)5

(C)3 (D)8

Q.17 The circuit shown below, is a controlled adder/half subtractor. The inputs to half adder/half subtractor are x and y while z is a control. The outputs are y_1 and y_2.

Then, the circuit behaves as



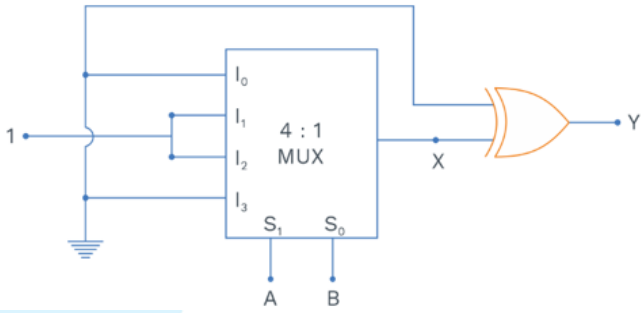
(A)Half subtractor regardless of whether z = 0 or z = 1 due to design.

(B)Half subtractor for z = 1 and half adder for z = 0.

(C)Half adder for z = 1 and half subtractor for z = 0.

(D)Half adder regardless of whether z = 0 or z = 1 due to design.

Q.18 The logic circuit shown in the figure below, the output (Y) is –



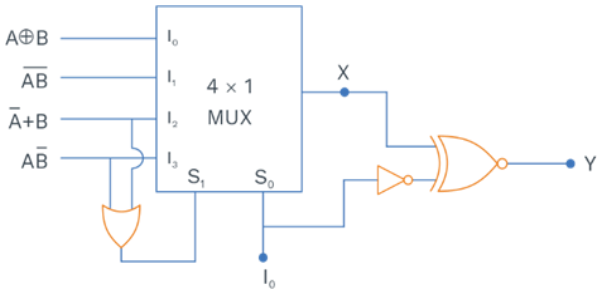
(A)$Y = \bar{A}B + \bar{A}\bar{B}$

(B)$Y = A \odot B$

(C)$Y = A \oplus B$

(D)Y = 0

Q.19 The Boolean function at the output of the given circuit is



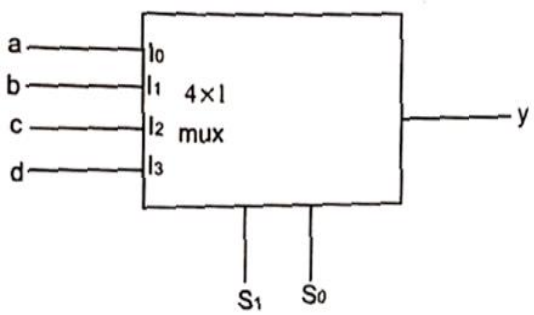
(A) Output $Y = \overline{A\bar{B}}$

(B) Output $X = \overline{A + \bar{B}}$

(C) Output $X = \overline{\bar{A}B}$

(D) Output $Y = \overline{\bar{A} + B}$

Q.20 Consider the following circuit



Where the multiplexer inputs a, b, c, d correspond to the respective segments of the 7-segment LED display as shown below.

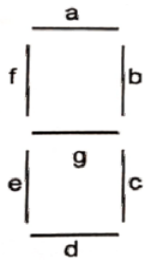


Diagram of 7 segment LED.

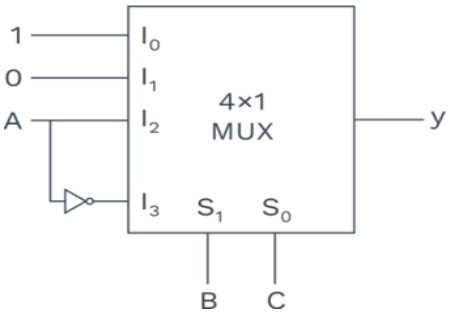
Consider a particular segment glowing (not glowing) as logic 1 (0).

The configuration of the switches S_1 and S_0 is same as digit displayed by the 7– segment display.

For example, if 3 is displayed, then $S_1 = 1$, $S_0 = 1$

For which of the following digits display the output "Y" is fixed at "0"?

Q.21 A 4×1 MUX is used to implement a 3-input Boolean function as shown in figure. The Boolean function y is –



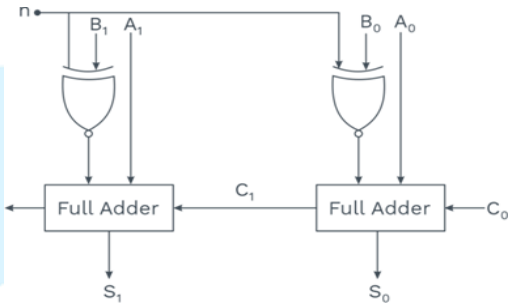
(A) $y = \sum m(0, 3, 4, 6)$

(B) $y = \pi M(0, 3, 4, 6)$

(C) $y = \Sigma m(1, 2, 5, 7)$

(D) $y = \pi M(1, 2, 5, 7)$

Q.22 The circuit is shown as



The operation that can be performed by the circuit if the control lines are selected accordingly are:

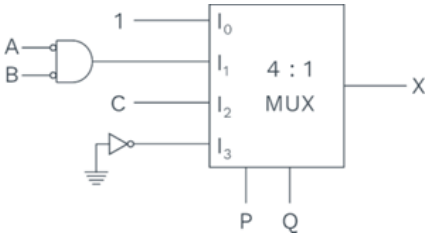
(A) A + B only

(B) A + B and A – B but not A + 1

(C) A + 1 and A + B but not A – B

(D) A + B, A – B and A + 1

Q.23 Consider the following circuit.



Which one of the following gives the function implemented by the MUX-based digital circuit?

(Where ground implies as logic "0")

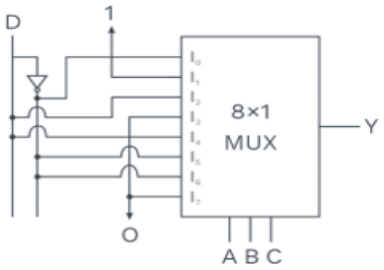
(A) $X = PQ + \bar{P}Q(\overline{AB}) + P\bar{Q}C + PQ$

(B) $X = (\overline{P+Q}) + \bar{P}Q(\overline{A+B}) + P\bar{Q}C + PQ$

(C) $X = (\overline{P+Q}) + \bar{P}Q(\overline{AB}) + P\bar{Q}C + \overline{PQ}$

(D) $X = (P+Q) + PQ(A+B) + PQC + \overline{PQ}$

Q.24 What is the expression of output (y) for the circuit shown in figure?



(A)Σm(2, 7, 9, 11, 12, 15)

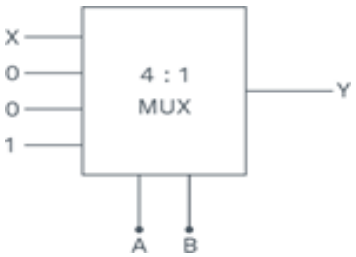
(B)πM(3, 8, 10, 13, 14)

(C)Σm(0, 2, 3, 5, 9, 10, 12

(D)πM(1, 4, 6, 7, 8, 11, 13, 14, 15)

Q.25 The minimum number of 2:4 decoders, required to implement a 3:8 decoder is _____.

Q.26 What should be connected at input X, so that the output of the given 4:1 MUX becomes Y=A·B ?



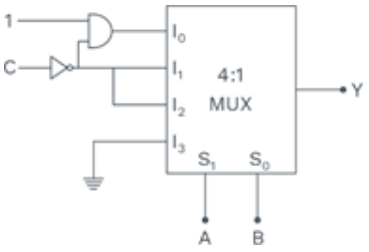
(A)Logic '1'

(B)Logic '0'

(C)A

(D)B

Q.27 A 4:1 MUX circuit is shown below, the minimized expression of output (Y) is:



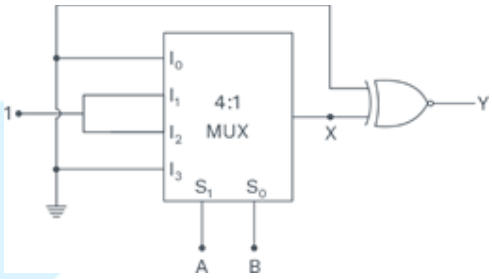
(A) $Y = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{C} + ABC$

(B) $Y = A \oplus B + \bar{B}\bar{C}$

(C) $Y = \bar{A}\bar{C} + \bar{B}\bar{C}$

(D) $Y = \bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + ABC$

Q.28 The logic circuit shown in the figure below, the output (y) is:



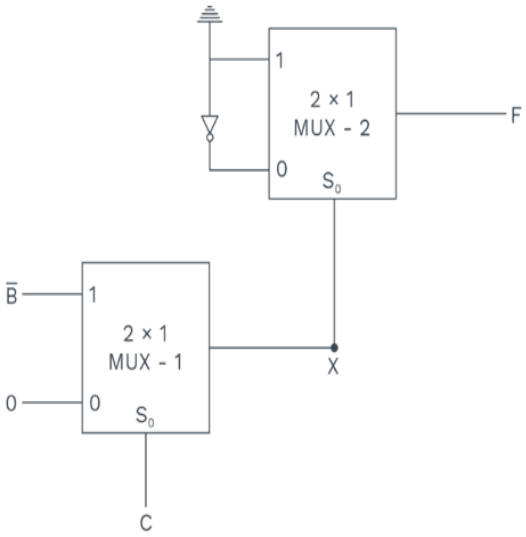
(A) $A \oplus B$

(B) $A \odot B$

(C)0

(D)1

Q.29 The logic circuit shown in the figure implements.



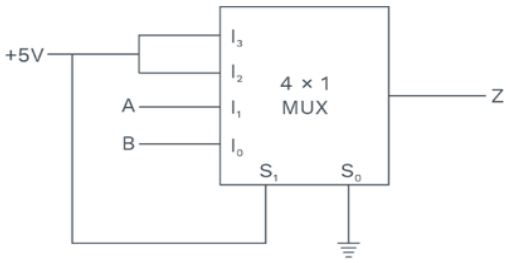
(A)F = 1

(B)F = 0

(C)$F = C\overline{B}$

(D)$F = \overline{C} + B$

Q.30 The MUX shown in the figure is 4 × 1 multiplexer. The output Z is –



(A)Z = A ⊕ B

(B)Z = A

(C)Z = 1

(D)Z = 0

Q.31 Consider a full adder where:

$A = 0, \quad B = 0$ and $C_{in} = 1$

and

$A = 1, \quad B = 0$ and $C_{in} = 1$

Then the values of Sum (S) and Carry (C_0) are-

(A)$S = 0, C_0 = 1 \& S = 1, C_0 = 1$

(B)$S = 0, C_0 = 0 \& S = 0, C_0 = 0$

(C)$S = 1, C_0 = 1 \& S = 1, C_0 = 0$

(D)$S = 1, C_0 = 0 \& S = 0, C_0 = 1$

Q.32 Consider the following statements:

1. Minimum number of NOR gate required to implement full adder is 8.

2. Minimum number of NOR gate required to implement half subtractor is 5.

3. Minimum number of NAND gate required to implement half adder is 5.

4. Minimum number of NAND gate required to implement NOR gate is 4.

Which of the above statements are correct?

(A)1, 2 and 3

(B)2, 3 and 4

(C)1, 3 and 4

(D)1, 2, 3 and 4

Q.33 How many 2-input multiplexers are needed to implement 512-input multiplexer and find the levels -

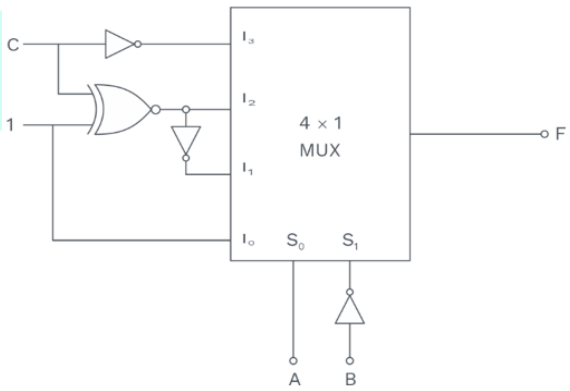
A)128, 9

(B)255, 8

(C)127, 8

(D)510, 9

Q.34 Circuit diagram of multiplexer is given below, find the output F –



(A)$F(A, B, C) = AB + \overline{A}C + AC$

(B)$F(A, B, C) = (A \odot C) + A\bar{B}$

(C)$F(A, B, C) = (A \oplus C) + \bar{A}B$

(D)$F(A, B, C) = \bar{A}\bar{C} + \bar{A}B + AC$

Q.35 The output Y of a 2-bit comparator is logic 1 whenever the 2-bit input A is greater than the 2-bit input B. The

number of combinations for which the output is logic 1, is

(A) 4

(C)

Q.36 In a half-subtractor circuit with X and Y as inputs, the Borrow (M) and Difference ($N = X - Y$) are given by

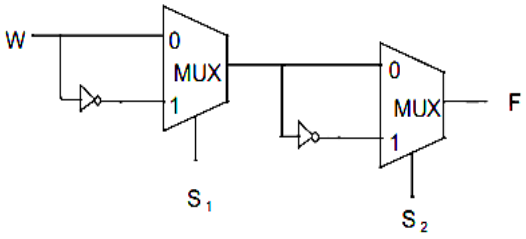
(A) $M = X \oplus Y$, $N = XY$

(B) $M = XY$, $N = X \oplus Y$

(C) $M = \bar{X}Y$, $N = X \oplus Y$

(D) $M = X\bar{Y}$, $N = \overline{X \oplus Y}$

Q.37 Consider the multiplexer based logic circuit shown in the figure.



Which one of the following Boolean functions is realized by the circuit?

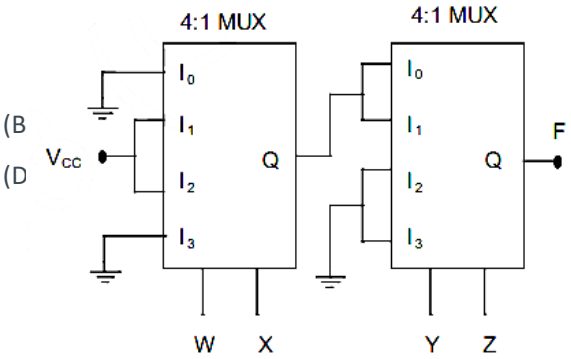
(a) $F = W\ \bar{S}_1\bar{S}_2$

(b) $F = WS_1 + WS_2 + S_1S_2$

(c) $F = \bar{W} + \bar{S}_1 + \bar{S}_2$

(d) $F = W \oplus S_1 + S_2$

Q.38 In the circuit shown, W and Y are MSBs of the control inputs. The output F is given by



a) $F = W\bar{X} + \bar{W}X + \bar{Y}\bar{Z}$

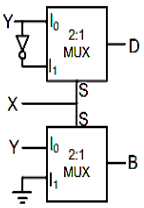
b) $F = W\bar{X} + \bar{W}X + \bar{Y}Z$

c) $F = W\bar{X}\bar{Y} + \bar{W}X\bar{Y}$

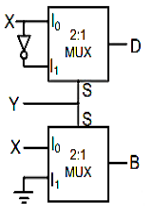
d) $F = (\bar{W} + \bar{X})\bar{Y}\bar{Z}$

Q.39 If **X** and **Y** are inputs and the Difference **(D = X – Y)** and the Borrow **(B)** are the outputs, which one of the following diagrams implements a half-subtractor?

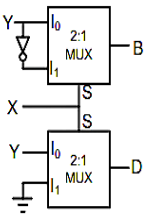
(A)



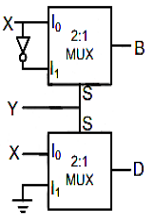
(B)



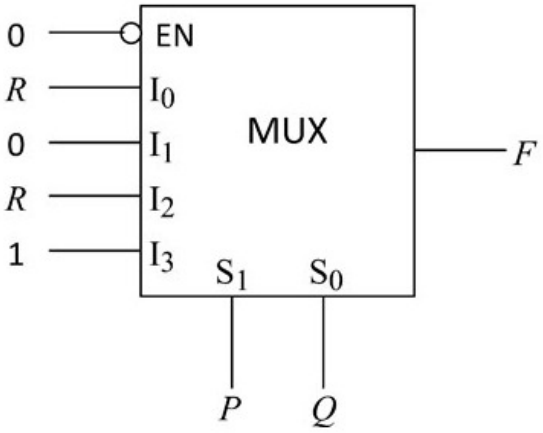
(C)



(D)



Q.40 The figure below shows a multiplexer where S_1 and S_0 are the select lines, I_0 to I_3 are the input data lines, EN is the enable line, and F(P,Q,R) is the output. F is



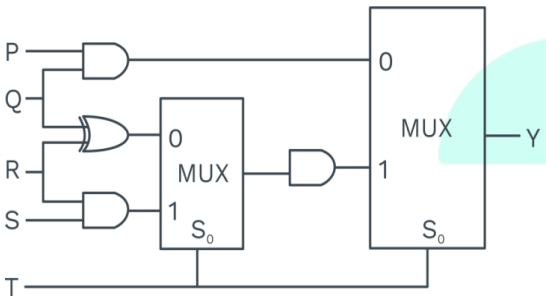
(A) $PQ + \overline{QR}$

(B) $P + Q\overline{R}$

(C) $P\overline{QR} + \overline{P}Q$

(D) $\overline{Q} + PR$

Q.41 The propagation delays of the XOR gate, AND gate and multiplexer (MUX) in the circuit shown in the figure are 4 ns, 2 ns and 1 ns, respectively.



If all the inputs P, Q, R, S and T are applied simultaneously and held constant, the maximum propagation delay of the circuit is

(A) 3 ns

(B) 5 ns

(C) 6 ns

(D) 7 ns

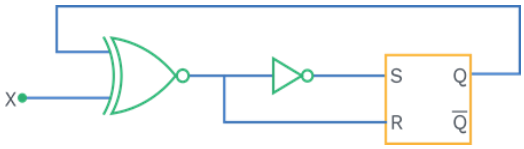
Answer Key

Q.1 Ans. [A ,B ,C]
Q.2 Ans. [B]
Q.3 Ans. [A ,B]
Q.4 Ans. [D]
Q.5 Ans. [C]
Q.6 Ans. [A ,D]
Q.7 Ans. [D]
Q.8 Ans. [A]
Q.9 Ans. [C]
Q.10 Ans. [5]
Q.11 Ans. [A ,D]
Q.12 Ans. [A]
Q.13 Ans. [A ,B ,C]
Q.14 Ans. [D]
Q.15 Ans. [2059]
Q.16 Ans. [B]
Q.17 Ans. [C]
Q.18 Ans. [C]
Q.19 Ans. [A ,C]
Q.20 Ans. [2]
Q.21 Ans. [A ,D]
Q.22 Ans. [B]
Q.23 Ans. [B]
Q.24 Ans. [C ,D]
Q.25 Ans. [2]
Q.26 Ans. [B]
Q.27 Ans. [C]
Q.28 Ans. [B]
Q.29 Ans. [D]
Q.30 Ans. [C]
Q.31 Ans. [D]
Q.32 Ans. [B]
Q.33 Ans. [D]
Q.34 Ans. [C]
Q.35 Ans. [B]
Q.36 Ans. [C]
Q.37 Ans. [D]
Q.38 Ans. [C]
Q.39 Ans. [C]
Q.40 Ans. [A]
Q.41 Ans. [C]

5 Sequential Circuits

MCQ, MSQ and NAT

Q.1 The modified circuit of an SR flipflop is given below. The characteristic equation is-



(A)$Q(t+1) = X \oplus Q$

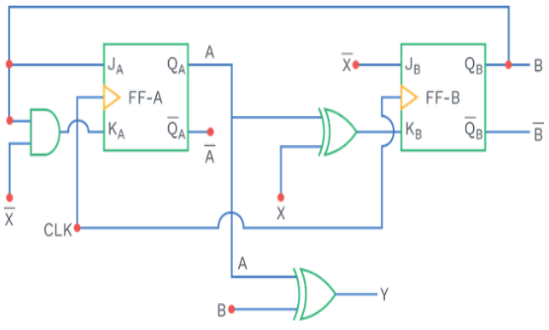
(B)$Q(t+1) = \overline{X \oplus Q}$

(C)$Q(t+1) = X + Q$

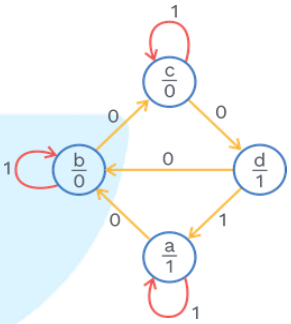
(D)$Q(t+1) = X$

Q.2 JK flipflops having a propagation delay of 15 nsec each, are used to construct the ripple counter. If the counter is operating at 8 MHz, then the largest MOD number of the counter is_____.

Q.3 A sequential circuit along with its state diagram is shown below.



If Y is considered as the output, then what will be the correct value of a, b, c, and d respectively, in the given state diagram?



(A)10, 11, 00, and 01

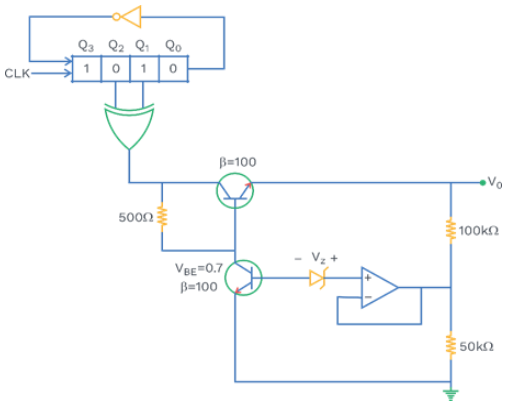
(B)00, 01, 10, and 11

(C)01, 10, 11, and 00

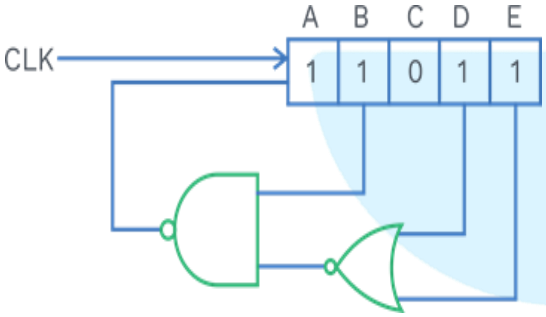
(D)11, 00, 01, and 10

Q.4 The shift register shown below is initially loaded with the bit pattern 1010. The pattern gets shifted by a one-bit position to the right for every clock pulse. If the Zener diode cut-off voltage is 5V, then the output voltage (V_0) after 3 clock pulses is _____ Volts. (Up to 1 decimal place)

[Assume logic '1' as 20V and logic '0' as 0 volts]



Q.5 A right shift register is initially set at 11011. Find the content of the shift register after 5 clock pulses.



(A)1 1 0 1 1

(B)1 1 1 1 1

(C)1 0 0 0 0

(D)0 0 1 1 1

Q.6 An AB flip-flop whose characteristic table is given below is implemented using an SR flip-flop.

A	B	Q+
0	0	0
0	1	1
1	0	1
1	1	0

Which one of the following statements is correct?

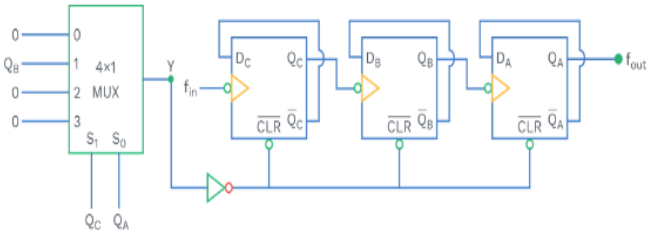
(A)$S = \overline{A + B}$ and $R = \overline{AB}$

(B)$S = A \oplus B$ and $R = A \odot B$

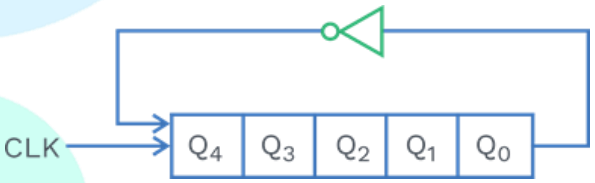
(C)$\mathrm{S = A \odot B}$ and $\mathrm{R = A \oplus B}$

(D)$S = \bar{A}B + A\bar{B}$ and $R = \bar{A}\bar{B}$

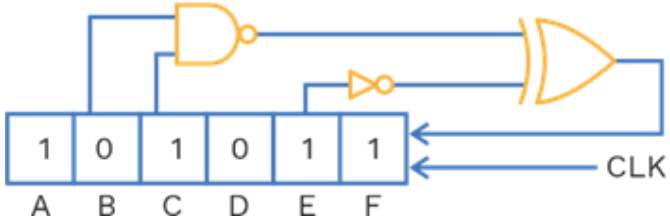
Q.7 For the given circuit, the output frequency (f_{out}) is 3.5 MHz. The input frequency (f_{in}) is _____ MHz.



Q.8 A Johnson counter circuit is given below. The number of unused states in the counter is _____.



Q.9 A left shift register is initially set at 101011. The content of the shift register after 6 clock pulses is-



(A)111001

(B)100110

(C)011011

(D)101101

Q.10 If the characteristic table of XY Flip-Flop given below is implemented using JK – Flip Flop, then-

X	Y	Q^+
0	0	1
0	1	0
1	0	1
1	1	1

(A)$J = X\bar{Q} + Y$

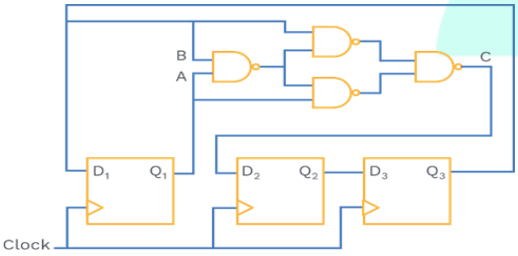
(B)$J = X + \bar{Y}$

(C)$K = \bar{X}Y$

(D)K = YQ

Q.11 A 5-bit counter-type ADC has a clock frequency of 3.5 MHz. The maximum conversion time is ------_____ µsec. (Rounded off to two decimal places)

Q.12 A circuit is shown in the figure below. Q_1 is MSB and Q_3 is LSB.



Which of the following statement(s) is/are true?

(A)The expression of C in terms of A and B is A ⊙ B

(B)Initially if $Q_1Q_2Q_3$ = 001, then after one clock pulse, $Q_1Q_2Q_3$ = 110

(C)The expression of C in terms of A and B is A ⊕ B

(D)Initially if $Q_1Q_2Q_3$ = 001, then after one clock pulse, $Q_1Q_2Q_3$ = 001

Q.13 The initial state of a MOD-8 down counter is 1110. After 51 clock pulses, the decimal value of the user-defined down counter will be_____. (Rounded off to the nearest integer)

Q.14 The difference between the number of unused states in the ring counter and the number of used states in the Johnson counter is_____. (Assume Number of bits = 5).

Q.15 A pulse train with a frequency of 8 MHz is counted using the MOD-512 ripple counter. If counter is designed using JK flipflops, then the maximum permissible propagation delay per flip flop stage is _____nsec. (Up to one decimal place).

Q.16 The characteristic table of an XY-Flipflop is given below.

X	Y	Q^+
0	0	1
0	1	1
1	0	1
1	1	0

If it is implemented using JK-flipflop, then-

(A)$J = X\bar{Q} + Y$

(B)$J = \bar{X} + \bar{Y}$

(C)K = XY

(D)K = YQ

Q.17 A synchronous counter is designed using two J-K flipflops with the following connections-

$J_0 = K_0 = \overline{Q}_1$, $J_1 = Q_0$ and $K_1 = \overline{Q}_0$

The circuit acts as a-

(A)Counter of MOD-2

(B)Counter of MOD-3

(C)Right shift register

(D)Left shift register

Q.18 The number of flip flops required to construct a MOD-31 counter is-

(A)4 (B)5

(C)3 (D)6

Q.19 Consider the following circuit of JK-flip flops and choose the correct statement.

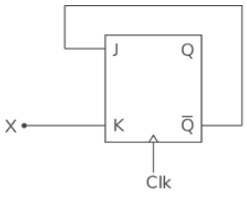


Figure (I)

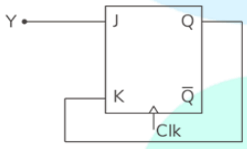


Figure (II)

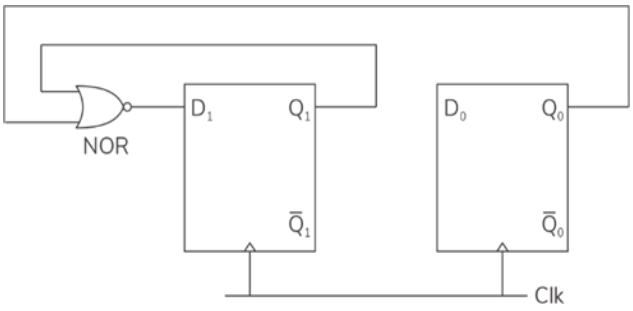
(A)(X = 1, Y = 1), (I) and (II) both behaves as inverter

(B)(X = 0, Y = 0), (I) and (II) both behaves as inverter

(C)(X = 1, Y = 0), (I) and (II) both behaves as inverter

(D)(X = 0, Y = 1), (I) and (II) both behaves as inverter

Q.20 In the sequential circuit shown below, what will be the state sequence of Q_1Q_0?



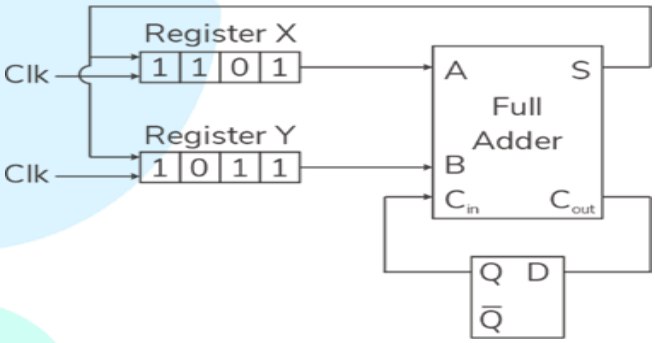
(A)00, 10, 01, 00 ...

(B)00, 11, 10, 00 ...

(C)00, 01, 10, 00 ...

(D)00, 10, 11, 00 ...

Q.21 What will be the content of register X after five clock pulses? (Assume the initial value of C_{in} = 0)



(A)1001

(B)1010

(C)0110

(D)110

Q.22 For the conversion of JK flip flop to D flip flop

(A)$J = D, K = \bar{D}$

(B)J = 0, K = D

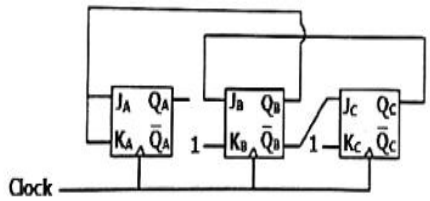
(C)J = D, K = D

(D)$J = \overline{D}, K = \overline{D}$

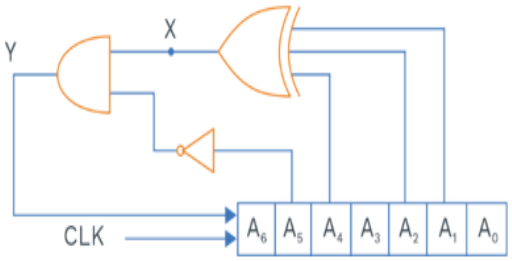
Q.23 An 8-bit ripple counter uses J-K flip-flops. If the propagation delay of each flip-flop is 200 ns, then the maximum

clock frequency that can be used is _____ kHz.

Q.24 Consider the sequential circuit given below. The number of unused states is_____ (in integer)



Q.25 The initial contents of right shift register shown below is 0101110, the content of the shift register after 5 clock pulses will be –



(A)0110010 and X = 0

(B)0011001 and X = 1

(C)0110010 and X = 1

(D)0011001 and X = 0

Q.26 A JK flip-flop has propagation delay of 15 nsec. What is the largest MOD number that can be constructed using flip-flops and operate upto 20MHz?

Q.27 The initial state of MOD-16 down counter is 1010. After 55 clock pulses output of the counter is

(A)0010

(B)0011

(C)0101

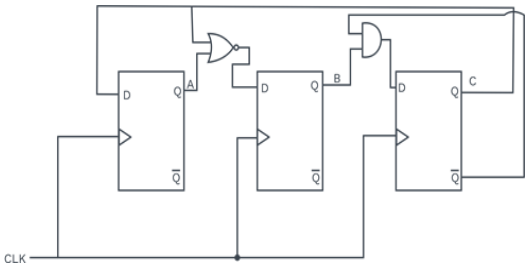
(D)1010

Q.28 A 4 – bit ripple counter uses flip flops with propagation delay time of 25 ns each and maximum possible time required for change of state is t_p. A 4 – bit synchronous counter uses flip flops with same propagation delay time and maximum possible time required for change of state is t_s.

Then $\frac{t_s}{t_p}$ is _____.

Q.29 A Mod-5 ripple counter uses flip flops having a delay of 40 nsec each. The clock pulse is applied when the output is taken. The maximum delay (in nsec) is _____.

Q.30 Consider the circuit involving 3 D-flip flops used in the counter configuration. Before the trigger, A, B and C has value 0, 1 and 0 respectively. The value of ABC after three clock edge is-



(A)011

(B)000

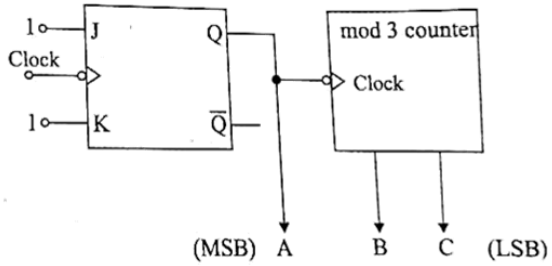
(C)11

(D)010

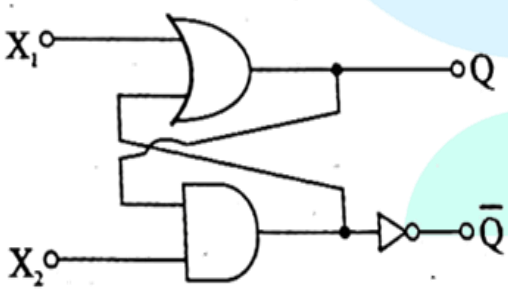
Q.31 A Mod-5 ripple counter uses flip-flops having a delay of 20 nsec each. The

clock pulse is applied when the output is taken. The maximum delay (in nsec) is _____

Q.32 In the figure shown below, initially ABC = 000. After three clock pulses, the decimal equivalent of state ABC will be



Q.33 The latching action of a latch shown below, is controlled by two inputs X_1, X_2. The invalid input combinational X_1, X_{-2} will be



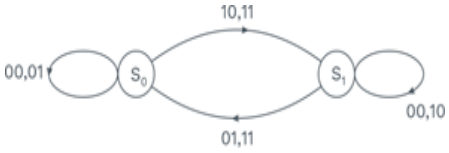
(A)$X_1 = 1, X_2 = 0$

(B)$X_1 = 0, X_2 = 1$

(C)$X_1 = X_2 = 0$

(D)$X_1 = X_2 = 1$

Q.34 The state diagram of the flip-flop is shown below. The type of flip-flop is:



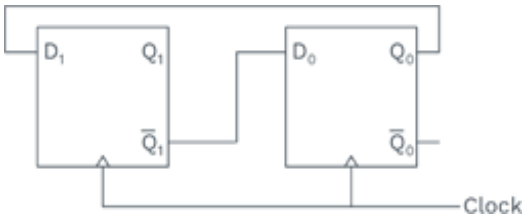
(A)SR Flip-flop

(B)T Flip-flop

(C)JK Flip-flop

(D)D Flip-flop

Q.35 The counter circuit shown below is initially cleared, find the state after 2 clock cycles.



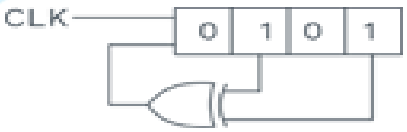
(A)$Q_1 = 0$, $Q_0 = 1$

(B)$Q_1 = 0$; $Q_0 = 0$

(C)$Q_1 = 1$; $Q_0 = 1$

(D)$Q_1 = 1$; $Q_0 = 0$

Q.36 The initial contents of the 4-bit SIPO, right shift register shown in the figure is 0101. After 3 clock pulses, the content of the shift register is



(A)1000

(B)0001

(C)1010

(D)0101

Q.37 A 4-bit modulo-16 ripple counter uses JK flip-flop. If the propagation delay of each FF is 50 ns, the maximum clock frequency is equal to

(A)2

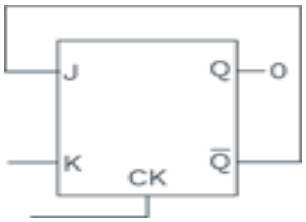
(B)3

(C)4

(D)5

Q.38 In a J-K flip-flop we have $J = \overline{Q}$ and $K = 1$ (shown in figure). Assuming

Assuming the flip-flop was initially cleared and then clocked for 6-pulses, the sequences at the $\overline{Q}_n$ output will be:



(A) 010000

(B)011001

(C)010010

(D) 010101

Q.39 Assume that the counter is holding the count 0100. What will be the count after 28 clock pulses?

(A)0101

(B)0001

(C)1111

(D)0000

Q.40 The number of unused states in a 4-bit Johnson counter is

(A)2

(B)4

(C)8

(D)12

Q.41 A 4-bit modulus 16 ripple counter uses a J-K flip-flop. If the propagation delay of each flip-flop is 100 nsec, the maximum clock frequency that can be used equals to -

(A)5 MHz

(B)2.5 MHz

(C)1.25 MHz

(D)10 MHz

Q.42 Assertion (A): Output of Mealy state machine is a function of the present state as well as present input.

Reason (R): Output of Moore state machine is a function of present state only.

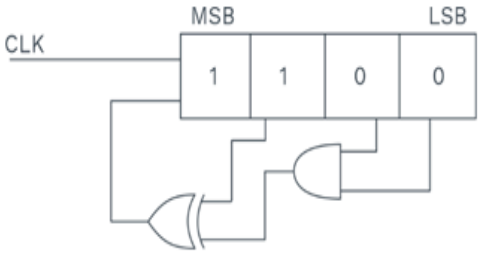
(A)Both assertion and reason are true, reason is the correct explanation of assertion

(B)Both assertion and reason are true, reason is not the correct explanation of the assertion

(C)The assertion is true, but the reason is false.

(D)The assertion is false, but the reason is true.

Q.43 A shift register is shown in the figure below.



The number of clock pulses required by the circuit to get the contents of the shift register as 1010 is-

(A)8

(B)7

(C)6

(D)5

Q.44 A 4-bit synchronous counter uses flip-flop with propagation delay of 50ns each. The maximum time required for the change of state is-

(A)50nsec

(B)100nsec

(C)200nsec

(D)400nsec

Q.45 A counter has 8 flip-flops which are initially at 00000000. The content of the flip-flops after 2156 clock pulses is-

(A)11011000

(B)10101100

(C)01101100

(D)11001001

Q.46 A 4-bit modulo-16 ripple counter uses a JK flip-flop. If the propagation delay of each FF is 75 ns, the maximum clock frequency is equal to

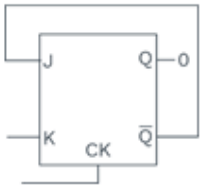
(A)75 MHz

(B)3.33 MHz

(C)1.67 MHz

(D)5 MHz

Q.47 In a J-K flip-flop we have $J = \overline{Q}$ and K = 1 (shown in the figure). Assuming the flip-flop was initially cleared and then clocked for 5-pulses, the sequences at the $\overline{Q_n}$ output will be:



(A)01000

(B)01100

(C)01001

(D)01010

Q.48 The number of unused states in a 3-bit Johnson counter is

(A)2

(B)4

(C)

(D)12

Q.49 A Mod-5 ripple counter uses flip flops having a delay of 20 nsec each. The clock pulse is applied when the output is taken. The maximum delay is-

(A)20nsec

(B)40nsec

(C)60nsec

(D)100nsec

Q.50 In serial in serial out (SISO) configuration, total number of clock pulses required to storing and retrieving the 10-bit data is -

(A)Total number of clock pulses required = 10

(B)Total number of clock pulses required = 1

(C)Total number of clock pulses required = 19

(D)Total number of clock pulses required = 16

Q.51 A 5-bit UP counter has stable states from 0000 to 1001, if the input frequency is 240 kHz then the output frequency of the counter is equal to -

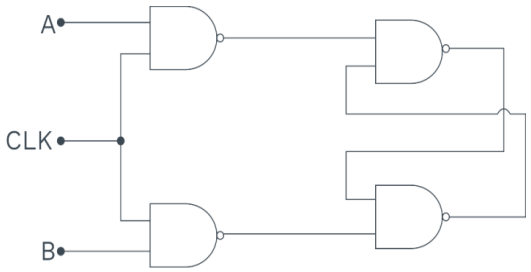
(A)50 kHz

(B)24 kHz

(C)12 kHz

(D)2.4 kHz

Q.52 Consider the given circuit.



In this circuit, the race around

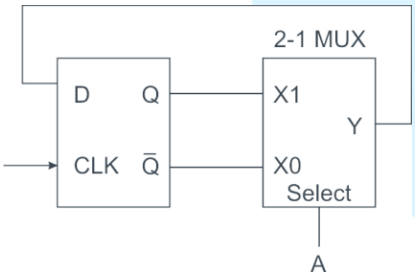
(A) does not occur

(B) occurs when CLK = 0

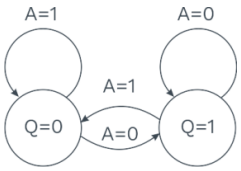
(C) occurs when CLK = 1 and A = B = 1

(D) occurs when CLK = 1 and A = B = 0

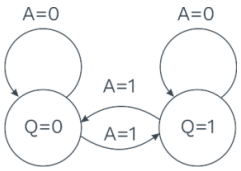
Q.53 The state transition diagram for the logic circular shown is



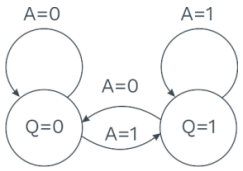
(A)



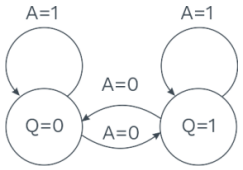
(B)



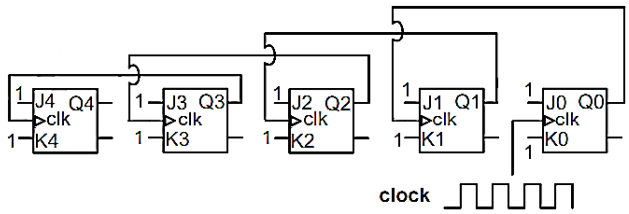
(C)



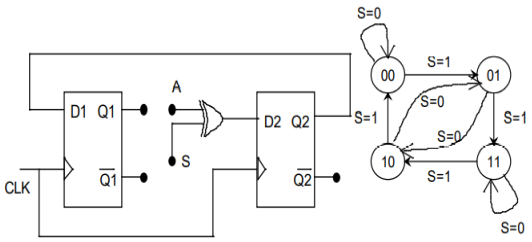
(D)



Q.54 Five JK flip-flops are cascaded to form the circuit shown in Figure. Clock pulses at a frequency of 1 MHz are applied as shown. The frequency (in kHz) of the waveform at Q3 is _____.



Q.55 The digital logic shown in the figure satisfies the given state diagram when Q1 is connected to input A of the XOR gate.



Suppose the XOR gate is replaced by an XNOR gate. Which one of the following options preserves the state diagram?

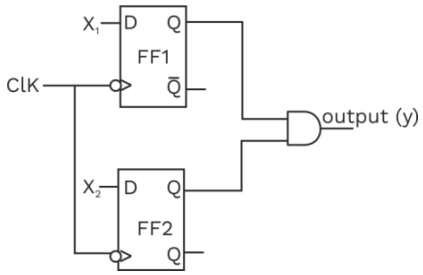
(A) Input A is connected to $\overline{Q2}$

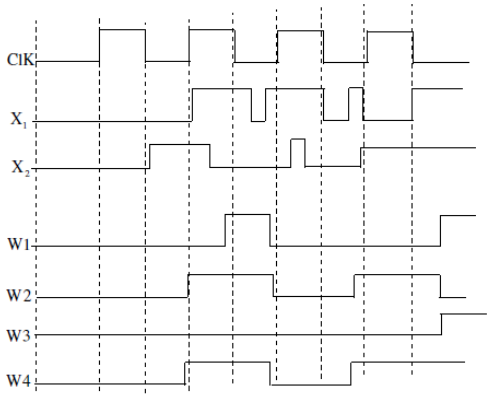
(B) Input A is connected to Q2

(C) Input A is connected to $\overline{Q1}$ and S is complemented

(D) Input A is connected to Q1

Q.56 In the circuit shown, choose the correct timing diagram of the output (y) from the given waveforms W1, W2, W3 and W4.

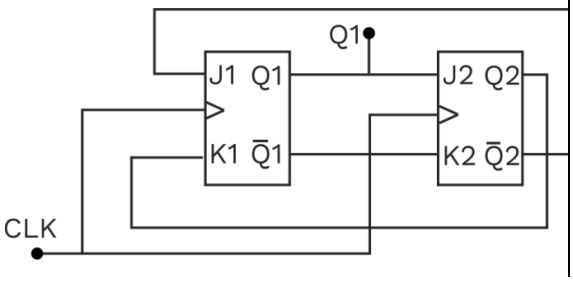




(A) W1 (B) W2

(C) W3 (D) W4

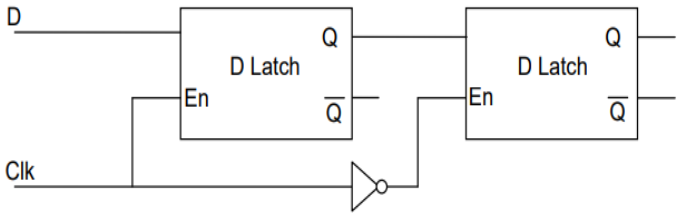
Q.57 The outputs of the two flip-flop Q1, Q2 in the figure shown are initialized to 0, 0. The sequence generated at Q1 upon application of clock signal is



(A) 01110.... (B) 01010....

(C) 00110.... (D) 01100....

Q.58 The circuit shown in the figure is a



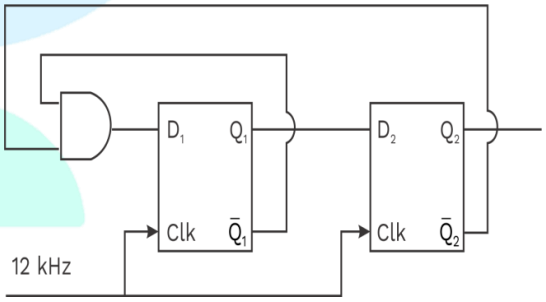
(A) Toggle Flip Flop

(B) JK Flip Flop

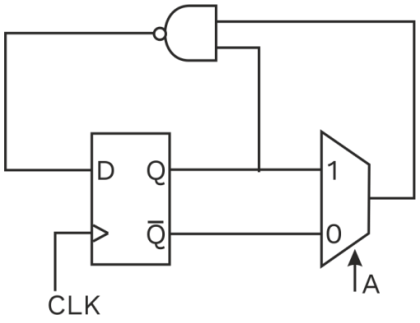
(C) SR Latch

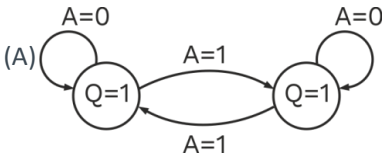
(D) Master-Slave D Flip Flop

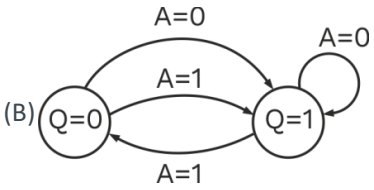
Q.59 In the circuit shown, the clock frequency, i.e., the frequency of the Clk signal, is 12 kHz. The frequency of the signal at Q_2 is _____ kHz.

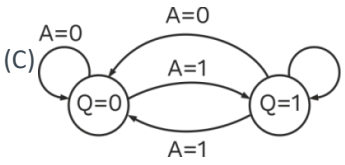


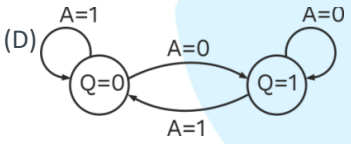
Q.60 The state transition diagram for the circuit shown is



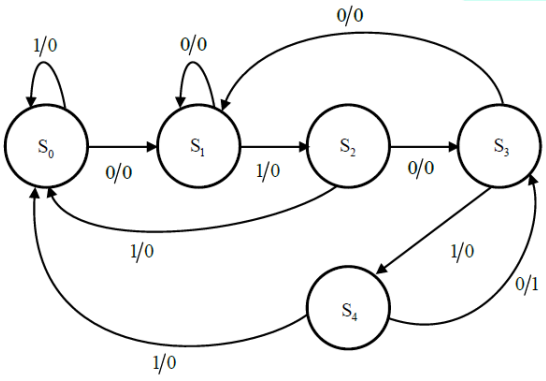








Q.61 The state diagram of a sequence detector is shown below. State S_0 is the initial state of the sequence detector. If the output is 1, then



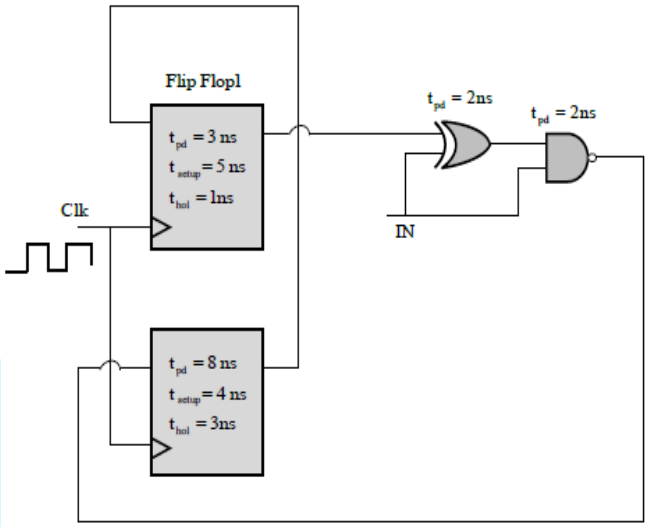
(A) the sequence 01010 is detected.

(B) the sequence 01011 is detected.

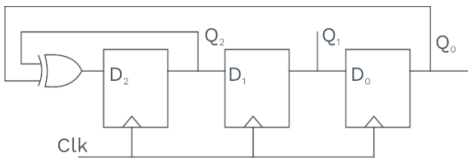
(C) the sequence 01110 is detected.

(D) the sequence 01001 is detected.

Q.62 For the components in the sequential circuit shown below, t_{pd} is the propagation delay, t_{setup} is the setup time, and t_{hold} is the hold time. The maximum clock frequency (rounded off to the nearest integer), at which the given circuit can operate reliably, is _____ MHz.



Q.63 The propagation delay of the exclusive-OR (XOR) gate in the circuit in the figure is 3 ns. The propagation delay of all the flip-flops is assumed to be zero. The clock (Clk) frequency provided to the circuit is 500 MHz.



Starting from the initial value of the flip-flop outputs $Q_2Q_1Q_0$= 111 with D_2= 1, the minimum number of triggering clock edges after which the flip-flop outputs $Q_2Q_1Q_0$becomes 100 (in integer) is _____ .

Answer Key

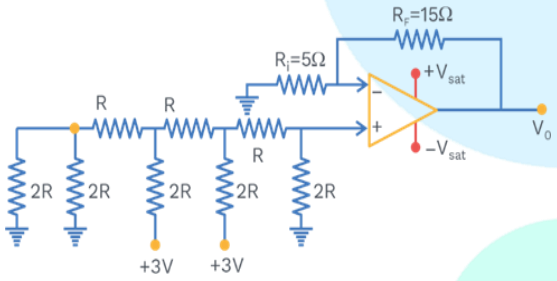
Q.1 Ans. [A]
Q.2 Ans. [256]
Q.3 Ans. [A]
Q.4 Ans. [17.1]
Q.5 Ans. [B]
Q.6 Ans. [B]
Q.7 Ans. [21]
Q.8 Ans. [22]
Q.9 Ans. [A]
Q.10 Ans. [B ,C]
Q.11 Ans. [8.86]
Q.12 Ans. [B ,C]
Q.13 Ans. [11]
Q.14 Ans. [17]
Q.15 Ans. [13.8]
Q.16 Ans. [B ,C]
Q.17 Ans. [B]
Q.18 Ans. [B]
Q.19 Ans. [A]
Q.20 Ans. [A]
Q.21 Ans. [D]
Q.22 Ans. [A]
Q.23 Ans. [625]
Q.24 Ans. [2]
Q.25 Ans. [B]
Q.26 Ans. [8]
Q.27 Ans. [B]
Q.28 Ans. [0.25]
Q.29 Ans. [120]
Q.30 Ans. [B]
Q.31 Ans. [60]
Q.32 Ans. [5]
Q.33 Ans. [A]
Q.34 Ans. [C]
Q.35 Ans. [C]
Q.36 Ans. [A]
Q.37 Ans. [D]
Q.38 Ans. [D]
Q.39 Ans [D]
Q.40 Ans. [C]
Q.41 Ans. [B]
Q.42 Ans. [B]
Q.43 Ans. [B]
Q.44 Ans. [A]
Q.45 Ans. [C]
Q.46 Ans. [B]
Q.47 Ans. [D]
Q.48 Ans. [A]
Q.49 Ans. [C]
Q.50 Ans. [C]
Q.51 Ans. [B]
Q.52 Ans. [A]
Q.53 Ans. [D]
Q.54 Ans. [62.5, Range 62.4 to 62.6)]
Q.55 Ans. [D]
Q.56 Ans. [C]
Q.57 Ans. [D]
Q.58 Ans. [D]
Q.59 Ans. [4]
Q.60 Ans. [B]
Q.61 Ans. [A]
Q.62 Ans. [76.92, Range is 76 to 77]
Q.63 Ans. [5]

6 DACs and ADCs

MCQ, MSQ and NAT

Q.1 An R-2R ladder type 4-bit DAC given below has a reference voltage of 18 volts. The output voltage V_0 (in volts) of the DAC is _____ (Rounded off to the nearest integer)

[Assume 0 V = logic '0' and + 3 V = logic '1']



Q.2 An 8-bit counter-type ADC has a clock frequency of 5MHz. The average conversion time for the ADC is _____ (in μsec) (Up to one decimal place)

Q.3 An 8-bit counter type ADC has maximum conversion time T_{c_1} and a 10-bit successive approximation type ADC has total conversion time T_{c_2}. If both ADCs have a clock frequency of 200 kHz, then $\frac{T_{c_2}}{T_{c_1}}$ is _____. (Up to three decimal places)

Q.4 An 8-bit DAC has a full-scale analog output voltage of 10 volts and the error is 0.25 % of full scale. Find the range of the analog output for the digital input 01101110.

(A)4.312 ± 0.025 volts

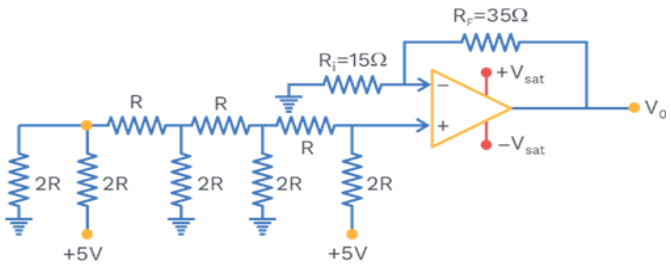
(B)1.632 ± 0.052 volts

(C)3.695 ± 0.52 volts

(D)6.562 ± 0.025 volts

Q.5 An R-2R ladder DAC given below has a reference voltage of 15 volts. Find the output voltage V_0 (in volts) of the DAC.

(Assume 0V = logic '0' and +5V = logic '1')



(A)3.265 V

(B)28.125 V

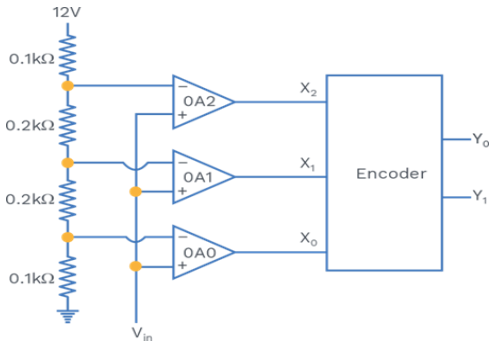
(C)18.726 V

(D)12.255 V

Q.6 The circuit shown below represents a 2-bit analog to digital converter

where the input voltage is $0 \leq V_{in} \leq 12$ Volts. The output $(Y_1Y_0)_{10}$ is _____.(Rounded off to the nearest integer)

Assume that the voltage $V_{in} = 8$ V.

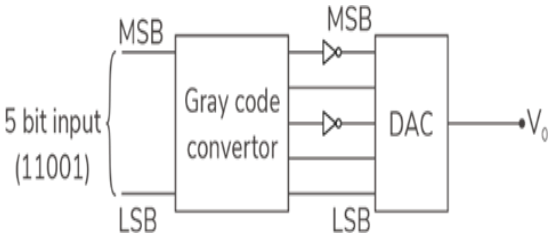


Q.7 The minimum number of bits used to construct a digital to analog converter, to keep the quantization error less than 0.4% of its full-scale range is _____.

Q.8 A 12-bit DAC that uses BCD input code has a full-scale output voltage of 19.98 V. What will be the output voltage of DAC for BCD input code of 0001 0101 0101?

[Up to 1 decimal place]

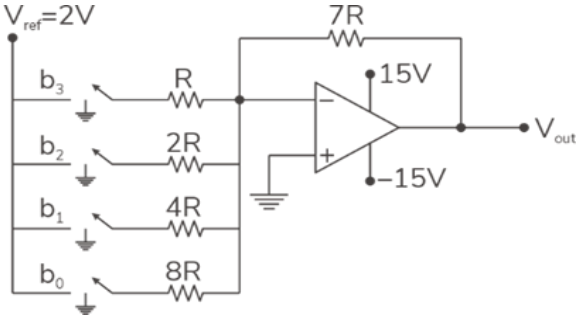
Q.9 Consider a 5-bit DAC circuit as shown below.



If full-scale voltage of DAC is 6.2 V, then the output voltage V_0 is_____.

(Up to 1 decimal place)

Q.10 Consider a weighted resistor type 'DAC' as shown below.



For proper functioning of DAC, what is the largest binary sequence that can be applied to the DAC?

(A)0110

(B)0111

(C)1010

(D)1111

Q.11 A 4-bit D/A converter gives an output voltage of 3.5 V for an input code of 0111. The output voltage for an input code of 0110 is _____ V.

Q.12 An 8-bit counter type ADC is operating at 3MHz clock frequency. The full-scale output voltage of ADC is 5.12 volts. The op-amp comparator used in the circuit has threshold voltage $V_T = 2$mV. The digital equivalent for input voltage $V_{in} = 1$V is

(A)$(01000111)_2$

(B)$(00101101)_2$

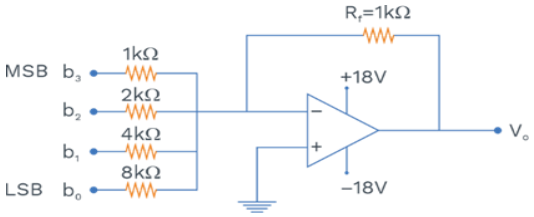
(C)$(10111011)_2$

(D)$(00110010)_2$

Q.13 Maximum conversion time (in msec) for a 10-bit digital ramp ADC, if the clock frequency is 30KHz, is _____.

(Up to 2 decimal places)

Q.14 In the 4-bit DAC binary weighted resistor shown below, the digital inputs are either 0V or 8V.



What is full scale output (in volts) of DAC for input 1111?

(A)–4.68

(B)–18

(C) –5.38

(D)–15

Q.15 The minimum no. of comparators required to build a 4-bit flash type A/D converter is

(A)16

(B)64

(C)15

(D)63

Q.16 An 8-bit D/A converter has a full-scale output voltage of 20V. The output voltage when the input is 11011011, will be

(A)160 mV

(B)78 mV

(C)20 V

(D)17 V

Q.17 What is the largest value of output voltage from a eight bit DAC that produces 5.0 V for a digital input of 00110010?

(A)5.1 V

(B)10.2 V

(C)2.55 V

(D)20.4 V

Q.18 There are two ADCs, 10-bit digital ramp ADC and a 10-bit successive approximation ADC. If both utilize a 500 kHz clock frequency, how much digital ramp faster than successive approximation?

(A)10 times

(B)20 times

(C)50 times

(D)100 times

Q.19 The resolution of a 16-bit analog to digital converter (in percentage) is -

(A)34.6

(B)0.00346

(C)15.2

(D)0.00152

Q.20 An n-bit A/D converter is required to convert analog input in the range 0-10 V to an accuracy of 20 mV, the value of n should be -

(A)10

(B)9

(C)8

(D)12

Q.21 An 8-bit D/A converter has a full-scale output voltage of 10V. The output voltage when the input is 11011011, will be

(A)160 mV

(B)78 mV

(C)20 V

(D)8.5 V

Q.22 The minimum number of bits used to construct digital to analog converter, to keep the quantization error less than 0.8% of its full-scale range is

(A)7 bits

(B)10 bits

(C)6 bits

(D)15 bits

Q.23 Maximum conversion time for a 12-bit digital ramp ADC if the clock frequency is 10 kHz -

(A)204.8 µsec

(B)409.6 µsec

(C)204.8 msec

(D)409.6 msec

Q.24 A 10-bit D/A converter is calibrated over the full range from 0 to 10 V. If the input to the D/A converter is 13A (in hex), the output (rounded off to three decimal places) is _____V.

Q.25 An 8-bit unipolar (all analog output values are positive) digital-to-analog converter (DAC) has a full-scale voltage range from 0 V to 7.68 V. If the digital input code is 10010110 (the leftmost bit is MSB), then the analog output voltage of the DAC (rounded off to one decimal place) is _____ V.

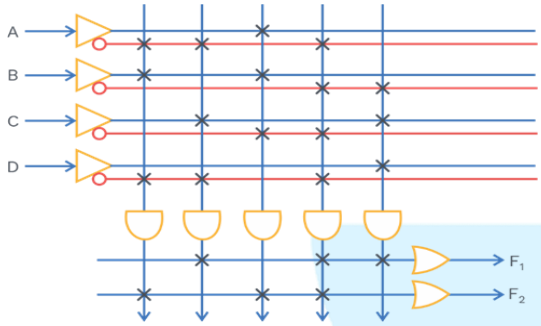
Answer Key

Q.1	Ans.	[27]
Q.2	Ans.	[25.6]
Q.3	Ans.	[0.039]
Q.4	Ans.	[A]
Q.5	Ans.	[B]
Q.6	Ans.	[2]
Q.7	Ans.	[8]
Q.8	Ans.	[3.1]
Q.9	Ans.	[0.2]
Q.10	Ans.	[B]
Q.11	Ans.	[3]
Q.12	Ans.	[D]
Q.13	Ans.	[34.13]
Q.14	Ans.	[D]
Q.15	Ans.	[C]
Q.16	Ans.	[D]
Q.17	Ans	[C]
Q.18	Ans	[D]
Q.19	Ans.	[D]
Q.20	Ans.	[B]
Q.21	Ans.	[D]
Q.22	Ans.	[A]
Q.23	Ans.	[D]
Q.24	Ans.	[3.066, Range 3.050 to 3.080]
Q.25	Ans.	[4.5]

7 CMOS Logic Family

MCQ, MSQ and NAT

Q.1 The simplified Boolean expressions for F_1 and F_2 of the given circuit are-



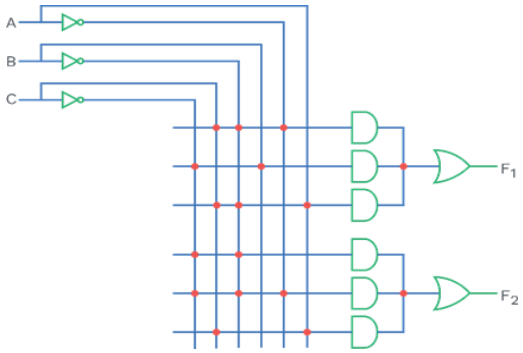
(A)$F_1 = \bar{A}\bar{B}\bar{D} + \bar{B}CD + \bar{A}C\bar{D}$

(B)$F_2 = AB\bar{C} + \bar{A}B\bar{D} + \bar{A}\bar{C}\bar{D}$

(C)$F_1 = \bar{A}\bar{B}CD + \bar{B}CD + \bar{A}BC\bar{D}$

(D)$F_2 = A\bar{B}\bar{C} + \bar{A}B\bar{D} + \bar{A}\bar{C}D$

Q.2 A programmable logic array circuit is given below. If the Schottky technology is used for the designing, the output(s) of the given circuit will be-



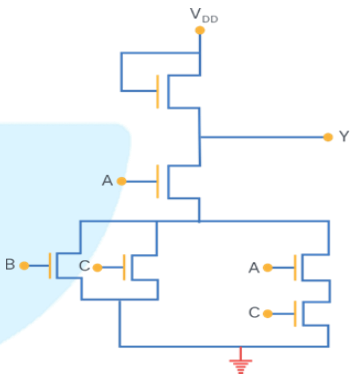
(A)$F_1 = B \odot C$

(B)$F_2 = A \oplus C$

(C)$F_2 = \bar{B}\bar{C} + AC$

(D)$F_1 = B \oplus C$

Q.3 Find the expression for Y in the given NMOS circuit.



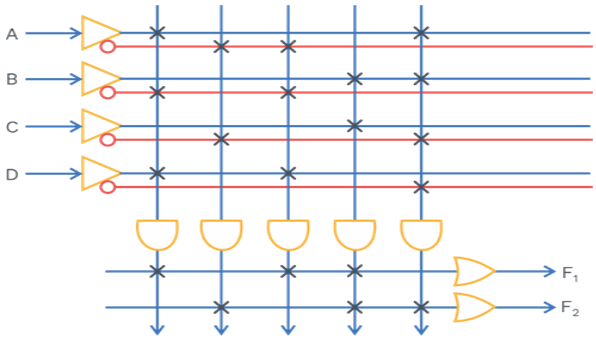
(A)$Y = \overline{AB + C}$

(B)Y = AB + AC

(C)$Y = \bar{A} + \bar{B}\bar{C}$

(D)$Y = AB + \bar{A}BC$

Q.4 Find the minterms (F_1 and F_2) of the given circuit.



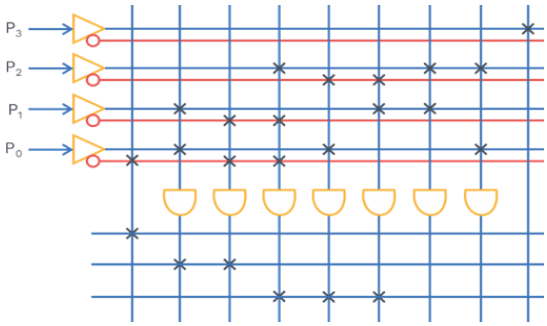
(A)$F_1 = \sum m(1, 2, 5, 7, 9, 10, 13, 15)$

(B)$F_2 = \sum m(0, 3, 4, 5, 7, 9, 10, 12, 14)$

(C) $F_1 = \sum m(1,3,6,7,9,11,14,15)$

(D) $F_2 = \sum m(0,1,4,5,6,7,12,14,15)$

Q.5 The given PLA circuit is a-



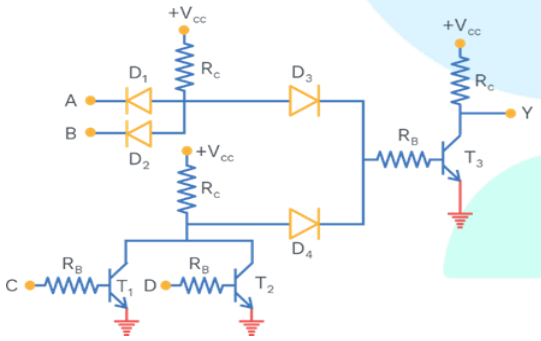
(A)Binary to gray code converter

(B)Gray to binary code converter

(C)BCD to Excess-3 code converter

(D)Excess-3 to BCD code converter

Q.6 The logical output (Y) of the given circuit is-



(A) $Y = \bar{A} + BC$

(B) $Y = \overline{A}BC + \bar{A}\bar{B}$

(C) $Y = \overline{AB + (C + D)}$

(D) $Y = \bar{A}C + \bar{A}D + \bar{B}C + \bar{B}D$

Q.7 Which one of the statements is **true** about the logic families?

(A)CMOS logic gate consumes more power than the ECL logic gate.

(B)ECL logic gate has the least propagation delay.

(C)I^2L logic gate has the highest speed of operation among all the logic gates.

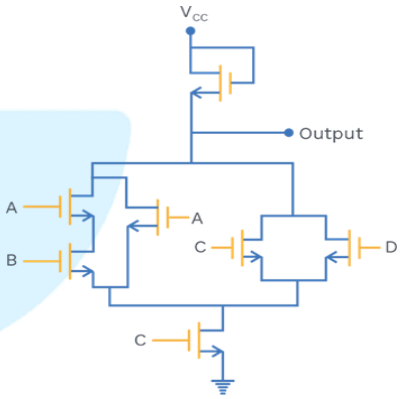
(D)RTL logic gate has the least fan-out.

Q.8 If a RAM chip has the size of (256 × 32) bits, then the number of RAM chips required to provide a memory capacity of 4096 Bytes is-

(A)32 (B)16

(C)4 (D)64

Q.9 A circuit is shown in the figure below.



If B = 0, D = 1 and the output is logic '1', then-

(A)A = 1 and C = 0 or 1

(B)A = 0 or 1 and C = 1

(C)A = 1 and C = 0

(D)A = 0 or 1 and C = 0

Q.10 Which of the following is/are not true about SRAM (Static RAM) and DRAM (Dynamic RAM)?

(A)SRAM performance is better than DRAM in terms of speed.

(B)SRAM performance is better than DRAM in terms of storage size.

(C)SRAM performance is better than DRAM in terms of access time

(D)SRAM performance is better than DRAM in terms of Ci (cost per bit).

Q.11 A ROM is used to implement Boolean functions given below.

$F_1(A,B,C,D) = \sum m(0,1,5,8) + \Sigma d(2,4,14)$

$F_2(A,B,C,D) = \Sigma m(1,6,7,8)$

What is the minimum size of the ROM required?

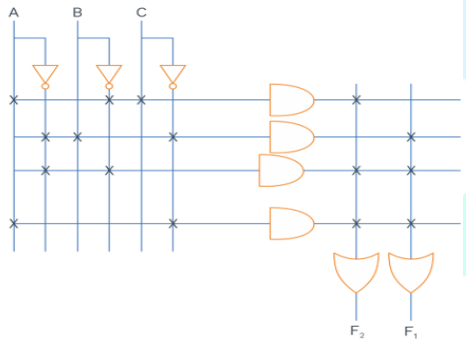
(A)32

(B)128

(C)64

(D)16

Q.12 Consider the PLA (Programmable logic array) circuit shown in the figure below. Find the minterm for outputs F_1 and F_2.



(A)$F_1 = \Sigma m(0,1,4,5,6)$ and $F_2 = \Sigma m(0,1,2,4,6)$

(B)$F_1 = \Sigma m(0,3,4,6)$ and $F_2 = \Sigma m(0,1,3,5,6)$

(C)$F_1 = \Sigma m(0,2,6)$ and $F_2 = \Sigma m(1,2,4,5,6)$

(D)$F_1 = \Sigma m(0,1,2,4,6)$ and $F_2 = \Sigma m(0,1,4,5,6)$

Q.13 How many 16K×1 RAM chips are needed to provide a memory capacity of 256 K- bytes?

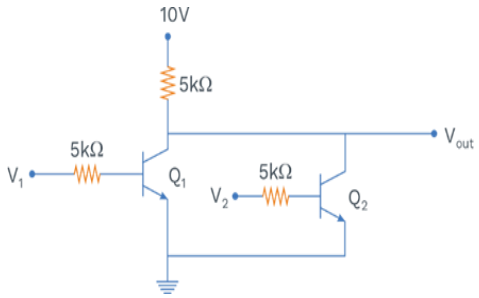
(A)128

(B)64

(C)256

(D)8

Q.14 A logical circuit given below has inputs V_1 and V_2. (logic '0' = 0 volt and logic '1' = 10V).



The given circuit represents-

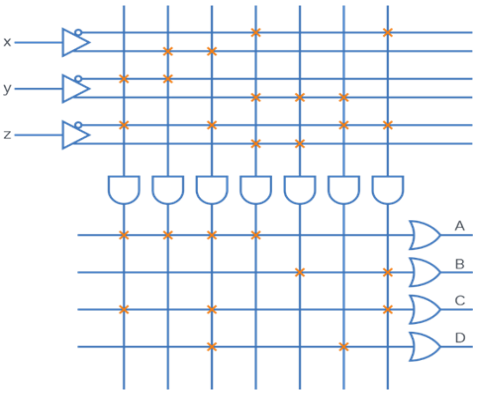
(A)NAND gate

(B)NOR gate

(C)NOT gate

(D)XOR gate

Q.15 Find the Boolean function for the given logical circuit.



(A)$C(x,y,z) = \Sigma(0,2,6,7)$ and $A(x,y,z) = \Sigma(0,3,4,5,6)$

(B)$A(x,y,z) = \Sigma(0,3,4,5,6)$ and $B(x,y,z) = \Sigma(0,2,3,7)$

(C)$\mathrm{D(x,y,z)} = \Sigma(2,4,6)$ and $\mathrm{C(x,y,z)} = \Sigma(0,2,4,6)$

(D)$A(x, y, z) = \Sigma(0, 3, 4, 5, 6)$ and $D(x, y, z) = \Sigma(2, 4, 6)$

Q.16 A ROM is used to implement Boolean function given below:

$F_1(A, B, C, D) = \Sigma m(0, 1, 5, 8) + \Sigma d(2, 4, 14)$

$F_2(A, B, C, D) = \Sigma m(1, 6, 7, 8)$

What is the minimum size of the ROM required?

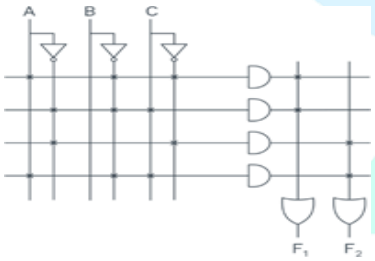
(A)32

(B)128

(C)64

(D)16

Q.17 Consider the PLA (Programmable Logic Array) circuit shown in the figure below, find minterms for output F_1 and F_2.



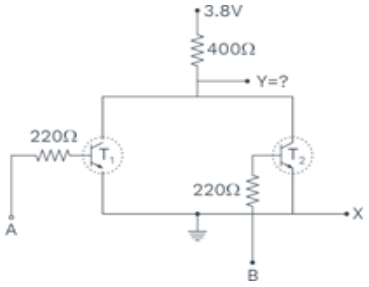
(A)$F_1 = \sum m(1, 4); F_2 = \sum m(0, 2, 5)$

(B)$F_1 = \sum m(0, 2, 6); F_2 = \sum m(2, 4)$

(C)$F_1 = \sum m(1, 3, 7); F_2 = \sum m(0, 2)$

(D)$F_1 = \sum m(1, 4); F_2 = \sum m(2, 4)$

Q.18 What will be the output Y of the given logic circuit.



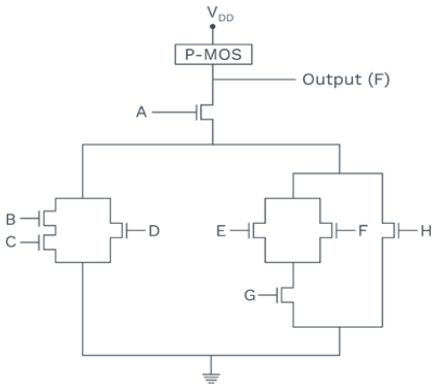
(A)$A \oplus B$

(B)$\overline{A + B}$

(C)A + B

(D)$\bar{A} + B$

Q.19 The expression for the circuit given below is-



(A)$\overline{A}[(\overline{B}\,\overline{C} + \overline{D}) + (((\overline{E} + \overline{F})\overline{G}) + \overline{H})]$

(B)$\overline{A[(BC + D) + (((E + F)G) + H)]}$

(C)$\bar{A}[(\bar{B} + \bar{C})\bar{D})(\bar{E}\bar{F} + \bar{G})\bar{H})]$

(D)$A + [(B + C)D)(EF + G)H)]$

Q.20 For the Boolean expressions given below, the ROM size required is:

$F_1(A, B, C) = \sum m(1, 2, 5, 6, 7)$

$F_2(A, B, C) = A\overline{C} + BA + \overline{B}C$

(A)$2^3 \times 2$

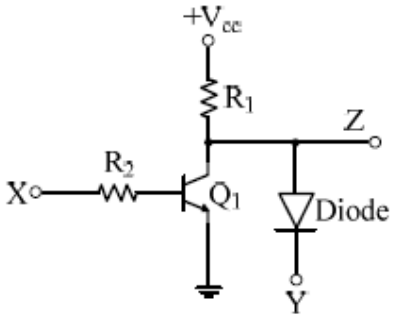
(B)$2^2 \times 3$

(C)$2^5 \times 3$

(D)$2^2 \times 2$

Q.21 In the circuit shown below, Q_1 has negligible collector-to-emitter

saturation voltage and the diode drops negligible voltage across it under forward bias. If V_{cc} is $+5V$, X and Y are digital signals with 0 V as logic and V_{cc} as logic 1, then the Boolean expression for Z is



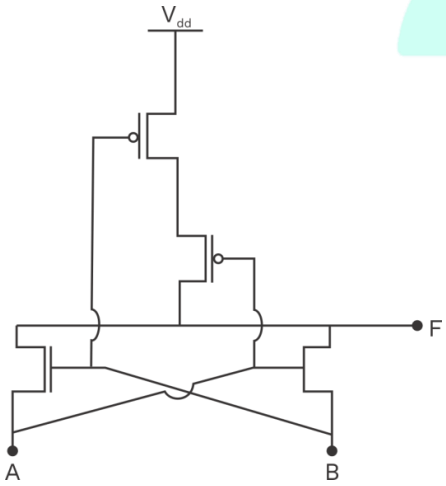
(A) XY

(B) $\overline{X}Y$

(C) $X\overline{Y}$

(D) $\overline{XY}$

Q.22 In the circuit shown, A and B are the inputs and F is the output. What is the functionality of the circuit?



(A) XOR

(B) XNOR

(C) Latch

(D) SRAM cell

Q.23 A standard CMOS inverter is designed with equal rise and fall times ($\tau_n = \tau_p$). If the width of the PMOS transistor in the inverter is increase, what would be the effect on the LOW noise margin (NM_L) and the HIGH noise margin NM_H?

(A) NM_L increases and NM_H decrease

(B) Both NM_L and NM_H increase

(C) No change in the noise margins

(D) NM_L decreases and NM_H increases

Answer Key

Q.1 Ans. [A ,B]

Q.2 Ans. [C ,D]

Q.3 Ans. [C]

Q.4 Ans. [C ,D]

Q.5 Ans. [C]

Q.6 Ans. [D]

Q.7 Ans. [B]

Q.8 Ans. [C]

Q.9 Ans. [D]

Q.10 Ans [B ,D]

Q.11 Ans. [A]

Q.12 Ans. [D]

Q.13 Ans [A]

Q.14 Ans. [B]

Q.15 Ans. [B ,C ,D]

Q.16 Ans. [A]

Q.17 Ans. [A]

Q.18 Ans. [B]

Q.19 Ans. [B]

Q.20 Ans [A]

Q.21 Ans. [B]

Q.22 Ans, [B]

Q.23 Ans. [A]

8 Computer Organization

MCQ, MSQ and NAT

Q.1 A non–pipelined system takes 200 nsec to process a task and the same task is processed by a 6-stage pipeline system in 45 nsec. The speedup of the pipeline system to the non-pipelined system for 250 tasks is_____ (Up to two decimal places)

Q.2 The addressing mode in which the value of the operand is present within the instruction itself is-

(A)Memory direct addressing mode

(B)Indexed addressing mode

(C)Immediate addressing mode

(D)Implicit addressing mode

Q.3 A disk drive has 6 surfaces, each surface is having 512 tracks, each track has 64 sectors, and each sector can hold 512 Byte. The rotation speed of the disk is 1500 RPM. The average rotational delay of the disk drive is_____ msec. (Rounded off to the nearest integer)

Q.4 A cache memory is designed using the write-through protocol. If the access time of the main memory is 250 nsec, the access time of the cache memory is 50 nsec, the hit rate of the read operation is 70% and the miss rate of the write operation is 60%, then the average access time of the read operation is _____ nsec. (Rounded off to the nearest integer)

Q.5 A drive has 8 surfaces, 8192 tracks per surface, and 256 sectors per track. If 64 bytes of data is stored in a sector using bit-serial manner, then-

(A)Number of sectors required = 2^{32} sectors

(B)Number of sectors required = 2^{24} sectors

(C)Width of sector address is 3 Bytes

(D)Width of sectors address is 4 Bytes

Q.6 Which of the statement(s) is/are true about the microprogramming?

(A)Vertical microprogramming is faster than horizontal microprogramming

(B)Horizontal microprogramming is faster than vertical microprogramming.

(C)Horizontal microprogramming requires signal decoder to decode the control signal.

(D)Vertical microprogramming requires signal decoder to decode the control signal.

Q.7 Assume that the RISC processor contains 40 global registers, 15 local registers, and the common register count is 8. If the processor contains 6 register windows, then the size of the window in the processor is_____.

Q.8 Consider a pipeline having 6 stages with cycle times of 4.2 nsec, 7.8 nsec, 3.6 nsec, 1.7 nsec, 5.5 nsec, and 2.6 nsec respectively. If the pipeline buffer time is 1.2 nsec, then the cycle time (in nsec) of the pipeline is_____ (Rounded off to the nearest integer)

Q.9 Consider a disk with R/W head. If the capacity of the track is 256 × 16 KB and the disk is rotating at 5000 rpm, then

(A)The capacity of the track = 2^{22} Byte

(B)Data transfer rate = 333.33 Mbps

(C)The capacity of the track = 2^{18} Byte

(D)Data transfer rate = 125.65 Mbps

Q.10 Assume that we have two pipelines P1 and P2 respectively. P1 has 8 stages having execution times of 14 ns, 36 ns, 13 ns, 16 ns, 18 ns, 5 ns, 15 ns, and 20 ns. P2 has 5 stages each having an execution time of 25 ns. If 200 instructions are to be executed, then the time saved by using the P2 pipeline over the P1 pipeline is_____nsec. (Rounded off to the nearest integer)

Q.11 The main memory unit is 32-bit wide and has a capacity of 1GB. The given memory is built using 256 × 4 Mbit DRAM chips. The number of rows in the chip is equal to 2^{15}. The percentage of time available with the main memory unit to perform the read/write operation is_____%. (Up to one decimal place)

(Assume that the time required by memory for one refresh operation is 40 nsec and the refresh period is 3 msec)

Q.12 Consider a pipeline having 5 stages with cycle times of 1.2 nsec, 5.6 nsec, 3.3 nsec, 8.7 nsec, and 2.6 nsec respectively. If the pipeline buffer time is 2.3 nsec, then the cycle time (in nsec) of the pipeline will be_____ (Rounded off to the nearest integer).

Q.13 Consider a system having 128 MB of memory. The word length of the CPU is 16 bits. What is the number of bits required to address each word of memory?

[Rounded off to the nearest integer]

Q.14 Consider a 9-stage instruction pipeline, all the states are perfectly balanced such that there is no cycle-time overhead of pipeline. The application is executing on 5-stage pipeline, and if 25% of the instruction incur 5 pipeline stall cycle, then the speed up achieved by with respect to non-pipelined execution is

(A)2 (B)4

(C)9 (D)5

Q.15 A RAM is having a capacity of 2K × 8. The number of 2×4 decoders required to construct 32K × 16 RAM from the given RAM of 2K × 8 is _____ . (in integer)

Q.16 Assume a processor in which the clock rate without pipelining is 10GHz. It has an average CPI of 10. If the processor is upgraded to pipelined processor with 5 stages, clock rate of the processor decreases to 3GHz due to internal delays. Assume that there is one stall cycle per instruction in pipelined processor. The speedup of pipelined processor over non-pipelined processor is -

(A)0.15

(B)0.52

(C)1.5

(D)2.52

Q.17 A pipeline is having 5 phases with duration of 20 nsec, 50 nsec, 30 nsec, 60 nsec, 10 nsec and 80 nsec. If latch delay is 30 nsec, then the speed-up ratio of pipelined over non-pipelined processor is _____.(up to 2 decimal places).

Q.18 How many 16 k×1 RAM chips are needed to provide a memory capacity of 256 kbytes?

(A)128

(B)64

(C)256 (D)8

Q.19 Consider a 9-stage instruction pipeline, all the states are perfectly balanced such that there is no cycle-time overhead of the pipeline. The application is executing on the 5-stage pipeline, and if 25% of the instruction incur 5 pipeline stall cycle, then the speedup achieved by with respect to non-pipelined execution is:

A)2 (B)4

(C)9 (D)5

Q.20 Suppose we are using an addressing mode, it has two main properties:

1. The operands are stored in the memory.
2. The address of the memory location in which operand is stored is given in the register which is already mentioned in the instruction.

Given below are various addressing modes. Which one among them is satisfying both the properties?

(A)Base indexed

(B)Register direct

(C)Register indirect

(D)Immediate

Q.21 Which of the following statement(s) about the properties of CISC architecture is not true?

(A)CISC machine instructions may have complex addressing modes, that can require more clock cycles.

(B)CISC control units are typically micro-programmed so that the instruction set can be more flexible.

(C)In the CISC instruction set, all arithmetic and logic instructions must be register-based.

(D)CISC architectures may perform better in network-centric applications than RISC architecture.

Q.22 A RISC processor has 308 registers. Each window has 4 input, 10 local and 4 output registers. The number of register windows is 16. The total number of global registers are

(A)84

(B)62

(C)68

(D)32

Q.23 Registers R1 and R2 contain data values 200 and 250 respectively in decimal, and the word length of the processor is 4 bytes. The effective address of the memory operand for the instructions “LOAD R5, 10 (R1, R2)” will be

(A)255

(B)240

(C)128

(D)460

Q.24 The flag which tells about the number of 1 bit resulting from arithmetic or logical operation?

(A)Parity flag

(B)Zero flag

(C)Auxiliary flag

(D)Carry flag

Q.25 Which of the following is true about pipeline latency.

(A)It is the time taken by all the instructions to be completed.

(B)It is the time taken for fetching all the instructions.

(C)It is the time taken by executing stage for all the instructions.

(D)It is the time taken by first instruction to complete all the stages.

Q.26 The instructions which are dependent on the source instruction can be separated using pipeline latency of source registers. This technique is known as

(A)Compiler scheduling

(B)Operand forwarding

(C)Register renaming

(D)Branch Prediction

Q.27 Which of the given statement(s) is/are true :

S1: Hardwired control unit is used in designing RISC architecture.

S2: Control signals are in encoded form in the Horizontal microprogramming Control unit.

S3: Only register to register arithmetic operations are used in RISC architecture

S4: Vertical microprogramming is slower than horizontal microprogramming in terms of speed.

(A)S1 and S2

(B)S1, S2 and S3

(C)S2 and S3

(D)S1, S3 and S4

Q.28 Consider a micro-programmed control unit. In this unit, the microinstructions are referred to in control memory using which of the following component?

(A)Memory Address Register (MAR)

(B)Memory Data Register (MDR)

(C)Program Counter (PC)

(D)Micro-programmed Counter

Q.29 Consider the following statements for the control unit.

S1: Operating speed of vertical microprogramming is higher than that of horizontal microprogramming.

S2: Horizontal microprogramming needs signal decoders, like vertical microprogramming.

Which of the above statements are true?

(A)Only S1 is true

(B)Only S2 is true

(C)Neither S1 nor S2 is true

(D)Both S1 and S2 are true

Q.30 A RAM is interface with the microprocessor using full decode scheme if the last and first memory of RAM is respectively 2FFFH and 1000H, then what will be the total memory range of RAM.

(A)2FFFH

(B)2000H

(C)0FFFH

(D)1FFFH

Q.31 The size of a memory chip is 512 x 4 bits. The number of chips required to make 2KB of memory is-

(A)1

(B)4

(C)8

(D)16

Q.32 Assume a read request takes 60 nsec on a cache miss and 10 nsec on a cache hit. While running a program, it is observed that 90% of the processor is read request results in a cache hit. The average memory access time is?

(A)50 nsec

(B)60 nsec

(C)55 nsec

(D)14.5 nsec

Q.33 Statements about the memory are given below –

1. Flash memory always provide byte-level eraser.

2. Synchronous DRAM moves data IN and OUT under the control of system clock.

3. In DRAM, the processor presents address and control levels to the memory

4. STT-RAM is a good candidate to replace or supplement DRAM for main memory.

Which of the following statements are correct?

(A)1 and 3

(B)2 and 3

(C)3 and 4

(D) 1, 2 and 3

Q.34 A RISC processor has 298 registers. Each window has 6 input, 9 local, and 4 output registers. The number of register windows is 16. The total number of global registers are

(A)58

(B)62

(C)68

(D)32

Q.35 Registers R1 and R2 contain data values 205 and 240 respectively in decimal, and the word length of the processor is 4 bytes. The effective address of the memory operand for the instructions “LOAD R5, 10 (R1, R2)” will be

(A)255

(B)440

(C)128

(D)455

Q.36 The flag which tells about the number of 1 is even or odd as the result of any arithmetic or logical operation?

(A)Parity flag

(B)Zero flag

(C)Auxiliary flag

(D)Carry flag

Q.37 Consider the instruction set architecture of a general-purpose machine. In this architecture, we have 50 control signals, out of which 15 are mutually exclusive and others are not. The number of bits required in the control word for microprogram will be at least.

(A)45

(B)35

(C)39

(D)51

Q.38 Consider the statements about addressing modes :

1. Implied address Mode: Operand is specified implicitly in the definition of instruction.

2. Immediate Addressing Mode: Operand is specified in the address field of instruction.

3. Register Addressing Mode: Operand is specified in registers that are in CPU.

4. Register Indirect Addressing Mode: Register specifies the address of the operand.

Which of the above statements are correct?

(A)2 and 3

(B)1 and 2

(C)1,3 and 4

(D)1,2, 3 and 4

Q.39 A RAM is interfaced with the microprocessor using a full decode scheme if the last and first memory of RAM is respectively 3000H and 2000H, then what will be the total memory range of RAM

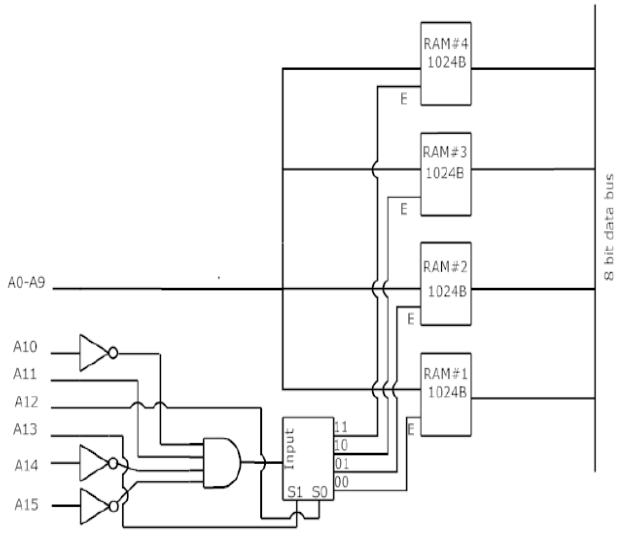
(A)2FFFH

(B)1001H

(C)0FFFH

(D)1FFFH

Q.40 There are four chips each of 1024 bytes connected to a 16 bit address bus as shown in the figure below. RAMs 1, 2, 3 and 4 respectively are mapped to addresses



(A) 0C00H-0FFFH, 1C00H-1FFFH, 2C00H-2FFFH, 3C00H-3FFFH

(B) 1800H-1FFFH, 2800H-2FFFH, 3800H-3FFFH, 4800H-4FFFH

(C) 0800H-0BFFH, 1800H-1BFFH, 2800H-2BFFH, 3800H-3BFFH

(D) 0800H-0BFFH, 1800H-1BFFH, 2800H-2BFFH, 3800H-3BFFH

Q.41 Addressing of a 32K × 16 memory is realized using a single decoder. The minimum number of AND gates required for the decoder is

(A) 2^8 (B) 2^{32}

(C) 2^{15} (D) 2^{19}

Answer Key

Q.1 Ans. [4.36]

Q.2 Ans. [C]

Q.3 Ans. [20]

Q.4 Ans. [125]

Q.5 Ans. [B ,C]

Q.6 Ans. [B ,D]

Q.7 Ans. [71]

Q.8 Ans. [9]

Q.9 Ans. [A ,B]

Q.10 Ans. [2352]

Q.11 Ans. [43.6]

Q.12 Ans. [11]

Q.13 Ans. [26]

Q.14 Ans. [B]

Q.15 Ans [5]

Q.16 Ans. [C]

Q.17 Ans. [2.27]

Q.18 Ans. [A]

Q.19 Ans. [B]

Q.20 Ans. [C]

Q.21 Ans. [C]

Q.22 Ans. [A]

Q.23 Ans. [D]

Q.24 Ans. [A]

Q.25 Ans. [D]

Q.26 Ans. [A]

Q.27 Ans. [D]

Q.28 Ans. [D]

Q.29 Ans [C]

Q.30 Ans [B]

Q.31 Ans. [C]

Q.32 Ans. [C]

Q.33 Ans. [B]

Q.34 Ans. [A]

Q.35 Ans. [D]

Q.36 Ans. [A]

Q.37 Ans. [C]

Q.38 Ans. [D]

Q.39 Ans. [B]

Q.40 Ans. D]

Q.41 Ans. [C]